

DETECTION OF BRAIN TUMOURS USING CONVOLUTIONAL NEURAL NETWORKS

A PROJECT REPORT

Submitted by,

Ms. K Seetha Thanooja - 20211CBD0054

Under the guidance of,

Mr. Ramamurthy ketha

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND TECHNOLOGY (BIG DATA)

At



PRESIDENCY UNIVERSITY

BENGALURU

JANUARY 2025

PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

CERTIFICATE

This is to certify that the Project report “**DETECTION OF BRAIN TUMOURS USING CONVOLUTIONAL NEURAL NETWORKS**” being submitted by “Seetha Thanooja Kalla” bearing roll number 20211CBD0054 in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Technology (Big Data) is a bonafide work carried out under my supervision.

Mr. Ramamurthy Ketha

Assistant Professor
School of CSE&IS
Presidency University

Dr. S. Pravinthraj

Professor & HoD
School of CSE&IS
Presidency University

Dr. L. SHAKKEERA

Associate Dean
School of CSE
Presidency University

Dr. MYDHILI NAIR

Associate Dean
School of CSE
Presidency University

Dr. SAMEERUDDIN KHAN

Pro-VC School of Engineering
Dean -School of CSE&IS
Presidency University

PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

DECLARATION

We hereby declare that the work, which is being presented in the project report entitled “**DETECTION OF BRAIN TUMOURS USING CONVOLUTIONAL NEURAL NETWORKS**” in partial fulfillment for the award of Degree of **Bachelor of Technology** in Computer Science and Technology (Big Data), is a record of our own investigations carried under the guidance of **Mr. Ramamurthy Ketha**, Assistant Professor, School of Computer Science Engineering & Information Science, Presidency University, Bengaluru.

We have not submitted the matter presented in this report anywhere for the award of any other Degree.

NAME

ROLL NUMBER

SIGNATURE

K. Seetha Thanooja

20211CBD0054

ABSTRACT

The growing incidence of brain tumors requires proper diagnostic tools for early detection and treatment. This project proposes a strong framework for the effective and high quality detection of brain tumors using Convolutional Neural Networks (CNNs), implemented through Google Colab and PyCharm. The proposed model uses advanced deep learning techniques to analyze Magnetic Resonance Imaging (MRI) scans, aiming to improve accuracy and efficiency in tumor identification. It starts by the collection of the comprehensive dataset MRI images that represent different types of brain tumors including meningioma, glioma, and pituitary. Preprocessing is of utmost importance where normalization, resizing, and data augmentation of images are carried out to make it more robust in the model.

The CNN model is trained on a labeled dataset, where features are extracted through convolutional layers that capture spatial hierarchies in the images. The training is done in Google Colab using its powerful GPU capabilities to process large datasets. PyCharm is used as the IDE for code development and testing, providing an opportunity to make iterative improvements and debug the code. Accuracy, sensitivity, specificity, and Area Under the Curve (AUC) are some of the metrics used to evaluate the performance of the model. The CNN implementation for the detection of brain tumors represents a great leap in medical imaging analysis. Using Google Colab for training and PyCharm for development streamlines the workflow so that researchers can focus on optimizing model parameters and enhancing performance. This automated detection system not only helps in timely diagnosis but also reduces the dependency on specialized radiological expertise, making it a valuable asset in clinical settings.

ACKNOWLEDGEMENT

First of all, we indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this project on time.

We express our sincere thanks to our respected dean **Dr. Md. Sameeruddin Khan**, Pro-VC, School of Engineering and Dean, School of Computer Science Engineering & Information Science, Presidency University for getting us permission to undergo the project.

We express our heartfelt gratitude to our beloved Associate Deans **Dr. Shakkeera L and Dr. Mydhili Nair**, School of Computer Science Engineering & Information Science, Presidency University, and **Dr. S. Pravinthraj**. Head of the Department, School of Computer Science Engineering & Information Science, Presidency University, for rendering timely help in completing this project successfully.

We are greatly indebted to our guide **Mr. Ramamurthy Ketha**, Assistant Professor and Reviewer Mr. Pakruddin B, School of Computer Science Engineering & Information Science, Presidency University for his inspirational guidance, and valuable suggestions and for providing us a chance to express our technical capabilities in every respect for the completion of the project work.

We would like to convey our gratitude and heartfelt thanks to the PIP2001 Capstone Project Coordinators **Dr. Sampath A K, Dr. Abdul Khadar and Mr. Md Zia Ur Rahman**, department Project Coordinators **Ms. Suma N.G** and Git hub coordinator **Mr. Muthuraj**.

We thank our family and friends for the strong support and inspiration they have provided us in bringing out this project.

K. Seetha Thanooja - 20211CBD0054

LIST OF TABLES

Sl. No.	Table Name	Table Caption	Page No.
1.	Table 1.1	System Requirements, Technical and Integration Specifications	4-5
2.	Table 1.2	Literature Survey	6-8

LIST OF FIGURES

Sl. No.	Figure Name	Caption	Page No.
1.	Figure 2.5	Performance Comparison of Different CNN Models.	21
2.	Figure 4.1	Model Workflow Architecture	31
3.	Figure 4.8	Confusion Matrix for the Dataset	33
4.	Figure 7.1	Gantt Chart	40
5.	Figure 9.1	Accuracy vs Validation Accuracy Graph	43
6.	Figure 9.2	Epoch Results	44

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iv
	ACKNOWLEDGMENT	v
	LIST OF TABLES	vi
	LIST OF FIGURES	vii
1.	INTRODUCTION	1
	1.1 Background	1-2
	1.2 Approach	2-3
	1.3 Problem Statement	3
	1.4 Requirement and Specifications	4-5
2.	LITERATURE SURVEY	6-8
	2.1 Historical Development	8
	2.1.1 Early Practices and Ancient Techniques	8
	2.1.2 Advent of Imaging Techniques.	9
	2.1.3 The Rise of Nuclear Imaging.	9
	2.1.4 The Emergence of Magnetic Resonance Imaging.	9-10
	2.1.5 Advances in Surgical Techniques.	10
	2.1.6 Integration of Artificial Intelligence.	10
	2.2 Imaging Techniques for Brain Tumor Detection.	10
	2.2.1 Conventional Magnetic Resonance Imaging (MRI)	10-11
	2.2.2. Computed Tomography (CT)	11-12
	2.2.3. Diffusion Tensor Imaging (DTI)	12
	2.2.4. Perfusion MRI	12
	2.2.5 Functional MRI (fMRI)	13
	2.2.6. Magnetic Resonance Spectroscopy (MRS)	13
	2.2.7 Tractography	13

2.2.8 Hybrid Imaging Techniques	14
2.2.9 Artificial Intelligence in Imaging.	14
2.3 Deep Learning in Medical Imaging.	14
2.3.1 Transfer Learning	14-15
2.3.2 Hybrid Models	15
2.3.3 Attention Mechanisms	15
2.3.4 Data Augmentation Techniques	16
2.3.5 Evaluation Metrics	16
2.4 CNN Architectures for Brain Tumor Detection	16
2.4.1 Common CNN Architectures for Brain Tumor Detection	16-17
2.4.2 Transfer Learning Approaches	17-18
2.4.3 Hybrid Models	18
2.4.4 Attention Mechanisms	18-19
2.5 Key Performance Metrics	19
2.5.1 Accuracy	19
2.5.2 Precision	19
2.5.3 Recall (Sensitivity)	19-20
2.5.4 F1 Score	20
2.5.5 Specificity	20-21
2.6 Future Directions and Research Opportunities in Brain Tumor Detection	21
2.6.1 Integration of Advanced Deep Learning Techniques	21-22
2.6.2 Transfer Learning and Pre-trained Models	22
2.6.3 Data Augmentation and Synthetic Data Generation	22-23
2.6.4 Real-Time Applications and Deployment	23
3. RESEARCH GAPS OF EXISTING METHODS	24
3.1 Data Limitations	24
3.1.1 Scarcity of High-Quality Datasets.	24
3.1.2 Class Imbalance	24
3.2 Challenges in Small Tumor Detection	24
3.2.1 Difficulty in Identifying Small Tumors	24

3.2.2 Proposed Solutions	
3.3 Model Performance Issues	25
3.3.1 Overfitting and Generalization	25
3.3.2 Evaluation Metrics	25
3.4 Interpretability and Explainability	25
3.4.1 Black Box Nature of Deep Learning Models	25-26
3.4.2 Proposed Solutions	26
3.5 Integration of Multi-Modal Data	26
3.5.1 Benefits of Multi-Modal Approaches	26
3.5.2 Proposed Solutions	26
3.6 Real-Time Detection Capabilities	26
3.6.1 Need for Real-Time Analysis	26
3.6.2 Proposed Solutions	27
3.7 Ensemble Learning Approaches	27
3.7.1 Benefits of Ensemble Methods	27
3.7.2 Proposed Solutions	27
3.8 Continuous Evaluation and Monitoring	27
3.8.1 Importance of Ongoing Evaluation	27
3.8.2 Proposed Solutions	28
3.9 Ethical Considerations in AI Deployment	28
3.9.1 Addressing Ethical Implications	28
3.9.2 Proposed Solutions	28
4. PROPOSED METHODOLOGY	29
4.1 Core Technologies Used	29
4.1.1 Python	29
4.1.2 Deep Learning	29
4.1.3 Custom-Based Convolutional Neural Networks (MobileNet Model)	29
4.1.4 HTML/CSS for Frontend Development	29-30
4.1.5 Google Colab and PyCharm	30
4.2 Methodology Overview	30

	4.3 Data Collection	
	4.3.1 Dataset Selection	30
	4.3.2 Data Annotation	30
	4.4 Data Preprocessing	30
	4.4.1 Image Resizing	30-31
	4.4.2 Normalization	31
	4.4.3 Data Augmentation	31
	4.5 Model Development	31
	4.5.1 Architecture Design	31
	4.5.2 Customization of MobileNet	32
	4.6 Training and Validation	32
	4.6.1 Splitting Data	32
	4.6.2 Model Training	32
	4.7 Evaluation Metrics	32-33
	4.8 Model Evaluation	33
	4.8.1 Testing on Unseen Data	33
	4.8.2 Confusion Matrix Analysis	33
	4.9 . Deployment	34
	4.9.1 Frontend Development	34
	9.2 Backend Integration	34
5.	OBJECTIVES	35
	5.1 Development of an Accurate Detection System	35
	5.2. Improvement of Diagnostic Capabilities	35
	5.3 Implementation of Advanced Image Processing Techniques	35
	5.4 Development of a User-Friendly Interface	35
	5.5 Evaluation and Validation of Model Performance	36
6.	SYSTEM DESIGN AND IMPLEMENTATION	37
	6.1 System Architecture	37
	6.1.1 Data Acquisition Module	37
	6.1.2 Preprocessing Module	37
	6.1.3 Model Development Module	37
	6.1.4 Training and Validation Module	38
	6.1.5 Evaluation Module	38
	6.2 Frontend Development	38
	6.2.1 User Interface Components	38
	6.3 Backend Development	38

	6.3.1 Framework Selection	
	6.3.2 API Development	39
	6.4 Deployment	39
	6.4.1 Cloud Deployment Options	39
	6.4.2 Continuous Monitoring	39
	6.5 Continuous Improvement Strategies	39
	6.5.1 Model Retraining	39
	6.5.2 Incorporation of User Feedback	39
7.	TIMELINE FOR EXECUTION OF PROJECT	40
	Gantt Chart	40
8.	OUTCOMES	41
	8.1 Enhanced Diagnostic Accuracy	41
	8.2 Effective Use of Transfer Learning	41
	8.3 User-Friendly Interface Development	41-42
	8.4 Contribution to Research and Knowledge Base	42
9.	RESULTS AND DISCUSSIONS	43-44
10.	CONCLUSION	45
11.	REFERENCES	46-47
12.	APPENDIX-A: PSUEDOCODE	48-58
13.	APPENDIX-B: SCREENSHOTS	60-63
14.	APPENDIX-C : ENCLOSURES	64

CHAPTER-1

INTRODUCTION

1.1 Background

The detection of brain tumors has become an area of critical research, since the condition is a serious health issue for human beings. Brain tumors are basically characterized by uncontrolled proliferation of abnormal cells; thus, they result in severe neurological disorders and require early diagnosis for timely treatment. Traditional methods of diagnosis depend entirely on the expertise of radiologists and may require much time and are prone to human error. To overcome these challenges, there has been a significant shift toward using advanced computational techniques, particularly Convolutional Neural Networks (CNNs), which have emerged as a leading approach in medical imaging analysis. CNNs are a class of deep learning algorithms specifically designed to process structured grid data, such as images. Their architecture mimics the human visual system, enabling them to automatically learn hierarchical feature representations from raw image data. CNNs are used for the detection of brain tumors.

They classify tumor types with a high degree of accuracy by analyzing MRI scans. It has been proved that CNN models can significantly outperform traditional methods in terms of the time taken to diagnose and increase the reliability of the results. Recent research work has focused on different architectures of CNNs, such as AlexNet and GoogLeNet. These models have been trained on huge datasets of MRI images, allowing them to learn the complex patterns that are associated with different types of brain tumors such as meningiomas, gliomas, and pituitary tumors. For example, it was shown in a study that GoogLeNet was able to achieve an accuracy of 99.45% in detecting brain tumors from MRI scans, beating AlexNet with an accuracy of 98.95%.

This is due to GoogLeNet's more efficient use of parameters and its deeper architecture, enabling it to capture more intricate features from images. Preprocessing techniques help in enhancing the performance of CNNs by normalizing and augmenting MRI images before feeding them into the model. This preprocessing is helpful in generating convolutional features that improve the model's ability to generalize across different datasets. The

evaluation metrics for these models are usually accuracy, sensitivity, specificity, and Area Under the Curve (AUC), providing a comprehensive view of their diagnostic capabilities. Applying CNNs into clinics is highly impressive for clinical fields in terms of medical images. Such models not only assist radiologists but also pave the way for more efficient healthcare delivery systems capable of improving patient outcomes through faster and more accurate diagnoses by automating the detection process.

1.2 Approach

Detection of brain tumors is a highly critical area of research because of the serious implications of this condition on human health. Brain tumors are basically characterized by uncontrolled growths of abnormal cells, which cause severe neurological disorders and require prompt diagnosis for proper treatment. The traditional methods of diagnosis are quite time-consuming and prone to human error, and they rely heavily on the expertise of radiologists.

To overcome these issues, there has been a strong trend toward the use of sophisticated computational methods, especially Convolutional Neural Networks (CNNs), which have become the front-runner in the analysis of medical images. CNNs are a type of deep learning algorithms tailored to process structured grid data, such as images. Their architecture is inspired by the human visual system, and they can learn hierarchical feature representations from raw image data automatically. In the context of brain tumor detection, CNNs analyze Magnetic Resonance Imaging (MRI) scans to identify and classify tumor types with high accuracy.

Studies have shown that CNN models can significantly outperform traditional methods by reducing the time required for diagnosis and increasing the reliability of results. Recent research has focused on various CNN architectures, including popular models like AlexNet and GoogLeNet. These models have been trained on large datasets of MRI images, enabling them to learn complicated patterns associated with different types of brain tumors, including meningiomas, gliomas, and pituitary tumors. For example, a study showed that GoogLeNet attained 99.45% accuracy in detecting brain tumors from MRI scans, which was better than that of AlexNet with an accuracy of 98.95%. This is because GoogLeNet uses parameters

more efficiently and is deeper than most other networks, allowing it to capture the finer features in images. Techniques in preprocessing can help improve the performance of CNNs through normalization and augmentation of MRI images before being fed into the model. It allows the model to generate better convolutional features and enhance its generalization ability over various datasets. These models are typically evaluated using metrics such as accuracy, sensitivity, specificity, and Area Under the Curve (AUC), providing a complete picture of their ability to diagnose.

Recent developments also include hybrid approaches that combine CNNs with other machine learning techniques or graph-based methods to improve classification accuracy further. For instance, some studies have proposed Graph Convolutional Networks (GCNs) that take into account non-Euclidean distances in image data, addressing limitations found in conventional CNN models. The integration of CNNs into clinical practice represents a significant advancement in the field of medical imaging. This, in turn, helps the models not only support radiologists but also opens doors to more effective health care delivery systems that can make a difference for patients by diagnosing them more rapidly and accurately.

1.3 Problem statement

The increased rate of brain tumors makes diagnosis timely and accurate extremely challenging, which is critical for proper treatment and good outcomes for patients. This is because most of the conventional methods rely on the knowledge of specialists such as radiologists, which tends to be very time-consuming and has room for errors by humans, thus causing unnecessary delays or misdiagnosis. This project attempts to address the above challenges through the development of an automated brain tumor detection system using Convolutional Neural Networks (CNNs) to analyze Magnetic Resonance Imaging (MRI) scans. The major objective is the development of a robust model to accurately identify and classify various types of brain tumors, such as gliomas, meningiomas, and pituitary tumors, while minimizing false positives and negatives. This project aims to improve diagnostic efficiency, reduce the burden on medical professionals, and ultimately improve patient care through faster and more reliable tumor detection by using deep learning techniques and advanced image processing methods.

1.4 Requirements and Specifications

Category	Specifications
Hardware Requirements	High-performance GPU (e.g., NVIDIA RTX 3090, Tesla K80, or similar)
	CPU with multiple cores (e.g., Intel i7/i9, AMD Ryzen 7/9)
	Minimum 16 GB RAM (32 GB or higher recommended for large datasets)
	Monitor with high resolution for visualizing MRI scans and model results
	Internet connection for downloading pre-trained models and libraries
Software Requirements	Operating System: Windows 10/11, Ubuntu 20.04+, or macOS.
	Programming Language: Python (3.7 or higher)
	Deep Learning Frameworks: TensorFlow, Keras, PyTorch
	Data Manipulation Tools: NumPy, pandas.
	Visualization Tools: Matplotlib, Seaborn.
Technical Considerations	Dataset: Preprocessed MRI images of brain tumors (e.g., BraTS dataset)
	Pre-trained Models: Access to models like ResNet, VGG, or EfficientNet for transfer learning
	Loss Functions: Cross-entropy for classification, Dice coefficient for segmentation
	Evaluation Metrics: Accuracy, Precision, Recall, F1-score, IoU (for segmentation tasks)
Integration Requirements	Web Application Frameworks: Flask or

	Django for deploying the model
	APIs: RESTful APIs for integrating the model into front-end interfaces
	Front-End: HTML5, CSS3, JavaScript for user interaction and visualization
	Security Measures: Data encryption, HIPAA compliance for handling sensitive medical data

Table - 1.1 System Requirements, Technical and Integration Specifications.

CHAPTER-2

LITERATURE SURVEY

AUTHOR(S)	YEAR	TITLE	SUMMARY
Akkus et al.	2017	"Deep Learning for Brain MRI Segmentation: State of the Art and Future Directions"	Proposed a CNN model for brain tumor segmentation in MRI scans. Achieved high accuracy for gliomas using 3D CNNs.
Swarup et al.	2023	Brain Tumor Detection Using CNN, AlexNet & GoogLeNet Ensembling Learning Approaches	Proposed a deep CNN model for automatic detection of brain tumors in MRI images, comparing AlexNet and GoogLeNet architectures. GoogLeNet achieved higher accuracy of 99.45%.
Bairagi et al.	2023	Automatic Brain Tumor Detection Using CNN Transfer Learning Approach.	Developed a CNN-based model utilizing transfer learning for brain tumor detection, achieving significant

			accuracy improvements.
Vandana et al.	2023	Brain Tumor Detection Using CNN with Data Augmentation	Implemented a CNN model with data augmentation techniques to enhance brain tumor detection accuracy in MRI images.
Balaji et al.	2022	Detection and Classification of Brain Tumors Using Deep Convolutional Neural Networks.	Employed various CNN architectures, including EfficientNetB0, ResNet50, and VGG16, for classifying glioma, meningioma, and pituitary tumors, achieving up to 97.61% accuracy.
Saxena et al.	2019	Predictive Modeling of Brain Tumor: A Deep Learning Approach	Utilized transfer learning with pre-trained models like ResNet-50 for binary classification of brain MRI scans, achieving 95% accuracy.
Chattopadhyay and Maitra	2022	MRI-Based Brain Tumor Image Detection Using	Suggested an automated method for multiclass

		CNN Based Deep Learning Method.	classification of brain tumors using a deep CNN model with six learnable layers.
Kushwaha and Maidamwar	2022	BTFCNN: Design of a Brain Tumor Classification Model Using Fused Convolutional Neural Networks	Designed a fused CNN model for brain tumor classification, enhancing accuracy through the integration of multiple CNN architectures.

2.1 Historical Development

2.1.1 Early Practices and Ancient Techniques

The history of brain surgery dates back to ancient civilizations, where rudimentary techniques were employed to treat head injuries and other ailments. One of the earliest known surgical procedures is trepanation, which involved drilling holes into the skull to relieve pressure or treat various conditions. Evidence of trepanation has been found in ancient cultures such as the Egyptians, Greeks, and Incas. These early procedures paved the way for subsequent neurosurgical treatments. Despite these early attempts at intervention, brain tumors were poorly understood. Not until the 19th century was real progress achieved in both surgical methodology and diagnosis.

The first well-documented primary brain tumor resection was completed by Rickman Godlee in 1884 and was an important milestone in neurosurgery. This operation presented the possibility for surgical intervention for the treatment of brain tumors, which opened up other developments in that field.

2.1.2 The Advent of Imaging Techniques

The discovery of Wilhelm Roentgen's X-ray in 1895 revolutionised medical imaging through the opening up of new diagnostics for brain tumor diagnosis. These early X-rays were not extremely useful in picturing soft structures such as a brain, however they were widely used to document calcified neoplasia and to achieve an understanding regarding cranial structures. Neurosurgeons like Harvey Cushing started using X-rays to localize tumors near the sella turcica, but their effectiveness was limited by the inherent limitations of the technology.

In 1919, Walter E. Dandy introduced ventriculography, a technique that involved removing cerebrospinal fluid from the brain's ventricles and replacing it with air to enhance X-ray visualization. This technique improved the localization of brain tumors but was demanding in terms of skills and experience to perform successfully. In the 1930s and 1940s, ventriculography emerged as a useful diagnostic tool, but later, more sophisticated imaging techniques took over.

2.1.3 The Rise of Nuclear Imaging

Nuclear medicine techniques introduced further progress in the mid-20th century for diagnosing brain tumors. During this time, the use of radioactive isotopes enabled the localization of tumours in the brain without invasion. In 1960, the research with nuclear scanning began and was able to reveal accurate identification of brain lesions. Through using isotopes like mercury-203, doctors were able to visualize the accumulation of tumors in certain parts of the brain. With this nuclear imaging advancing, these had been the early tools for seeing tumor characteristics and behavior. At that time though, they remained relatively less in comparison to the computed tomography and magnetic resonance imaging soon after.

2.1.4 The Emergence of Magnetic Resonance Imaging

MRI, one of the biggest advances in neuroimaging technology, was first discovered in the 1980s. It works on a principle where a strong magnetic field and radio waves produce images with very high resolution, with no use of ionizing radiation, and this could clearly differentiate between normal tissue and tumors, offering the best detail to date in brain

structures. In 1977, two scientists, Peter Mansfield and Paul Lauterbur, published the first human MRI images and were eventually awarded Nobel Prizes for their discoveries related to MRI technology. In the mid-1980s, MRI had already emerged as the standard reference technique in detecting brain tumors, given its sensitivity in demonstrating small tumors and estimating surrounding edema.

2.1.5 Advances in Surgical Techniques

As imaging technologies advanced, so did surgical techniques for the treatment of brain tumors. The late 20th century saw major advances in neurosurgery, especially with the development of microneurosurgery. This was developed by surgeons like Theodore Kurze and Mahmut Gazi Yaşargil, who used operating microscopes to improve the accuracy of tumor resections. Surgeons could thus better navigate complex anatomical structures and avoid damaging healthy tissue. Electrocautery devices further improved the results of surgeries by minimizing bleeding during operations.

2.1.6 Integration of Artificial Intelligence

There has been a recent surge of interest in the integration of artificial intelligence into brain tumor detection methodologies. The development of Convolutional Neural Networks (CNNs) has revolutionized medical image analysis by allowing for the automated classification of MRI scans. CNNs are designed to recognize intricate patterns within images, making them particularly effective for diagnosing various types of brain tumors. Recent studies have shown that CNNs can achieve diagnostic accuracies greater than 99%, which is higher than traditional methods that rely solely on human interpretation.

2.2 Imaging Techniques for Brain Tumor Detection

2.2.1 Conventional Magnetic Resonance Imaging (MRI)

MRI Technology Magnetic Resonance Imaging (MRI) is a non-invasive imaging technique that uses strong magnetic fields and radio waves to generate detailed images of soft tissues in the body, including the brain. MRI is particularly valuable for brain tumor detection due to its ability to provide high-resolution images without exposing patients to ionizing radiation. **MRI Protocols for Brain Tumors** Standard MRI protocols for brain

tumor evaluation typically include several sequences:

- T1-weighted imaging: It helps with anatomical details and detecting tumors.
- T2-weighted imaging: It increases edema and other abnormalities of tumors.
- Fluid-attenuated inversion recovery (FLAIR): Cerebrospinal fluid signal is suppressed, and this is helpful for enhancing the lesions' visualization.
- Diffusion-weighted imaging (DWI): Measures the diffusion of water molecules within tissues; in many instances, it gives a clue of cellularity within tumors.
- Gadolinium-enhanced sequences: The visualization of the blood-brain barrier disruption often occurs with tumors.

Advantages and Limitations MRI is the gold standard for brain tumor detection because of its excellent soft tissue contrast and ability to visualize complex anatomical structures. However, it has limitations, including longer scan times compared to CT scans and challenges in differentiating between tumor types based solely on imaging characteristics.

2.2.2. Computed Tomography (CT)

CT Imaging Computed Tomography, or CT, is an X-ray technique that produces images of the body in cross-section. This modality allows for rapid imaging, which is essential in emergency departments where rapid diagnosis is crucial. **CT Applications in Brain Tumors** CT scans are applied for:

- Initial evaluation of presumptive brain tumours.
- Assessing acute changes like hemorrhage or edema.
- Guiding biopsies or surgical planning.

Advantages and Limitations CT is faster than MRI and may be more available in emergencies. However, it exposes patients to ionizing radiation and has a lower sensitivity to

detect small lesions than MRI. The characterization of soft tissue details is poor with CT, which is crucial for the correct diagnosis of a tumor.

2.2.3. Diffusion Tensor Imaging (DTI)

Introduction to DTI Diffusion tensor imaging (DTI) is an advanced MRI technique that examines the diffusion of water molecules along white matter tracts in the brain. It provides information regarding the integrity of neural pathways and can be useful in assessing the effects of a tumor on adjacent brain structures. Applications of DTI in Brain Tumors DTI is especially useful in the following cases:

- Mapping white matter tracts before surgery to avoid damage during tumor resection.
- Measurement of alterations to white matter integrity due to infiltrating tumor tissue.

Advantages and Limitations

DTI allows valuable information that helps in a better planning for interventions on patients. It makes use of such information about structural connectivity of the brain, advanced processing techniques with the possible limited availability.

2.2.4. Perfusion MRI

Understanding Perfusion MRI Perfusion MRI assesses blood flow within the brain tissue, providing insights into tumor vascularity. This technique can help differentiate between high-grade and low-grade tumors based on their blood supply. Clinical Applications Perfusion MRI is particularly useful for:

- Grading tumors based on their vascular characteristics.
- Monitoring treatment response by evaluating changes in blood flow over time.

Advantages and Limitations Perfusion MRI can provide critical information about tumor biology that conventional imaging cannot offer. However, it may require specialized software for analysis and interpretation.

2.2.5 Functional MRI (fMRI)

Overview of fMRI Technology Functional MRI (fMRI) measures changes in blood oxygenation levels associated with neural activity. It helps assess functional areas of the brain before surgical interventions. **Applications in Brain Tumor Detection** fMRI is used for:

- Identifying eloquent cortex areas near tumors.
- Functional area-related surgery planning about regions linked with language, motor activity, and processing senses.

Advantages and Limitations fMRI provides real-time insights into brain function, aiding surgical planning. However, it is primarily a research tool and may not be routinely used in clinical practice for tumor detection.

2.2.6. Magnetic Resonance Spectroscopy (MRS)

Principles of MRS Magnetic Resonance Spectroscopy (MRS) completes the conventional MRI by giving the metabolic information regarding tissues. This technique analyses the chemical compounds within the brain to differentiate types of tumors based on metabolic profiles. **Applications of MRS in Brain Tumors** MRS can aid in Characterizing tumor metabolism by assessing levels of choline, creatine, and N-acetylaspartate. Tumor recurrence versus radiation necrosis based on metabolic signatures.

Advantages and Limitations MRS provides biochemical information that is not evident on conventional MRI scans, which enhances the accuracy of diagnosis. However, the interpretation of MRS is complicated and requires special expertise.

2.2.7 Tractography

Tractography is a visualization technique based on DTI data, which maps white matter pathways within the brain. It helps in visualizing how tumors may affect the surrounding neural tracts. **Clinical Applications** Tractography is very useful for Surgical planning by identifying critical pathways that need preservation during resection and also Post-operative change in white matter integrity.

2.2.8 Hybrid Imaging Techniques

Overview of Hybrid Techniques Hybrid imaging combines two different modalities for improved diagnostic accuracy. For example, PET/MRI combines the metabolic information obtained from PET with anatomical details from MRI. Applications in Brain Tumor Detection Hybrid techniques provide comprehensive data on both structure and function and may aid in:

- Accurate tumor characterization.
- Improved treatment planning through integrated assessments.

2.2.9 Artificial Intelligence in Imaging

It has shown promise for enhancing the diagnostic accuracy of imaging techniques by the integration of artificial intelligence. AI algorithms can look through large datasets of MRI scans and recognize patterns indicative of tumors. Applications of AI Algorithms AI applications include:

- Automatic segmentation of tumors from MR images.
- Deep learning models such as Convolutional Neural Networks are used to classify tumor types.

2.3 Deep Learning in Medical Imaging

2.3.1 Transfer Learning

Transfer learning is the process of taking a pre-trained model, originally trained on a large dataset, and fine-tuning it on a smaller, task-specific dataset. This is particularly useful in medical imaging where labeled data can be scarce. In brain tumor detection, transfer learning has been effectively applied using models such as:

- InceptionV3 and InceptionResNetV2: The models have been fine-tuned on MRI datasets to achieve very high accuracy for tumor classification tasks.

- MobileNetV2: It is a lightweight model that has been adapted for real-time tumor detection with great results, which is ideal for mobile applications.

2.3.2 Hybrid Models

Combining CNNs with Other Techniques Hybrid models integrate CNNs with other machine learning algorithms or neural network architectures to enhance performance:

- CNN-SVM Hybrid: Combining CNN feature extraction with Support Vector Machines (SVM) for classification can improve accuracy by leveraging SVM's robustness against overfitting.
- Ensemble Methods: Use multiple models and aggregate the predictions of them, which results in better classification performance by variance reduction and bias.

2.3.3 Attention Mechanisms

The Importance of Attention Mechanisms: Attention mechanisms allow models to pay attention to certain parts of an image that are more relevant to making predictions. This is very useful in medical imaging where the tumors may occupy only a small region of an MRI scan.

Convolutional Block Attention Module-CBAM The module extends CNN structures with both the application of spatial attention and that of channel wise:

- Channel Attention: This mechanism emphasizes important feature channels while suppressing less relevant ones.
- Spatial Attention: It pays attention to parts of the feature maps that are very sensitive for correct identification of tumors.

Researchers have achieved significant improvements in tumor detection accuracy by incorporating CBAM into models like YOLOv7.

2.3.4 Data Augmentation Techniques

Data augmentation involves artificially increasing the size of the training dataset through application of transformation steps like rotation, scaling, flipping, and adjustment of intensities. Such methods prevent overfitting and generally improve generalization in a model. For example, brain tumor detection:

- Augmentation techniques have been applied to improve the resolution of MRI images.
- The specific augmentation strategies can be selected according to the unique characteristics of different tumor types.

2.3.5 Evaluation Metrics

To evaluate the performance of deep learning models in the detection of brain tumors, the following evaluation metrics are used:

- Accuracy: The number of correctly classified instances over the total number of instances.
- Precision: The ratio of true positive predictions to the total number of positive predictions; it reflects how many selected items are relevant.
- Recall (Sensitivity): The ratio of true positive predictions to the actual number of positives; it reflects how well the model identifies positive cases.
- F1 Score: The harmonic means of precision and recall; it provides a balance between precision and recall.
- Area Under the Curve (AUC): A metric used to evaluate binary classification performance across different thresholds.

2.4 CNN Architectures for Brain Tumor Detection

2.4.1 Common CNN Architectures for Brain Tumor Detection

One widely utilized architecture in tasks related to the biomedical image segment is U-Net. In addition to encoding-decoding nature, it carries skip connections across both the paths for the information to be forwarded spatially. Use in Detection of Brain Tumors From MRI Scans U-Net has been extensively applied for identifying brain tumors. This has effectively helped in detailed localization and clear definition of boundary of the tumour.

Its architecture allows the model to capture both local and global features, thus enhancing its capability to identify tumors. Performance Metrics Studies using U-Net for brain tumor segmentation have achieved very high accuracy rates, sometimes greater than 95%. The architecture's capability of handling variable tumor sizes and shapes makes it more preferred for clinical applications.

Residual Learning Framework ResNet Residual learning is a new concept that uses skip connections allowing the gradients to flow more easily during training. This architecture corrects the problem related to vanishing gradients, a common problem deep networks face. **Implementation in Brain Tumor Detection** ResNet variants, such as ResNet50 and ResNet101, are used to classify brain tumors with MRI images. It has a deeper network which helps learn rich feature representations well. **Results and Accuracy Experiments** on classification of brain tumors have demonstrated successful classification results in a variety of cases using architectures ResNet. Results ranged between 96% to 99% based on the dataset as well as pre-processing techniques.

DenseNet connects each layer to every other layer in a feed-forward fashion, which encourages feature reuse and improves gradient flow throughout the network. **Application in Classifying Brain Tumors** DenseNet has been applied in various studies for classifying different types of brain tumors from MRI scans, leveraging its ability to capture intricate patterns in the data. **Performance Outcomes** DenseNet models have demonstrated high performance metrics, with accuracies often surpassing 97%. With an efficient use of parameters, they can maintain a competitive performance even with smaller datasets.

2.4.2 Transfer Learning Approaches

Transfer learning is about making use of models that are already pre-trained over large data and fine-tune them over smaller datasets, using limited amounts of data. **Pre-trained Models for Detection of Brain Tumor** Several pre-trained models are modified for detection of brain tumors:

- **VGG16/VGG19:** These are models that have simplicity and effectiveness for the image classification tasks.
- **InceptionV3:** This architecture uses inception modules that enable multi-scale feature extraction.
- **EfficientNet:** It reduces the model size while preserving the high accuracy with compound

scaling methods.

2.4.3 Hybrid Models

Hybrid models combine CNNs with other machine learning algorithms or neural network architectures to improve performance

- **CNN-SVM Hybrid:** Use feature extraction via CNN and supplement with Support Vector Machines for higher classification accuracy because SVM is less susceptible to overfitting.
- **Ensemble Learning Approaches:** Use multiple models to combine output, which could eventually lead to higher classification accuracy because it reduces variance and bias.

Case Studies on Hybrid Models Research demonstrated that hybrid models can achieve highly significant improvements in accuracy compared with standalone CNN architectures:

- A study of the combination of CNNs and SVM reported the accuracy increase by up to 5% more than traditional approaches using CNNs only.
- Multiple CNN architectures as ensemble methods are more robust with noisy data.

2.4.4 Attention Mechanisms

Attention mechanisms enable the model to pay attention to the parts of the image that are more relevant to making predictions. This is very helpful in medical imaging, for example, when a tumor may occupy only a small region of an MRI scan. CBAM enhances CNN architectures by integrating both channel-wise and spatial attention:

- **Channel Attention:** It highlights important channels of features while suppressing less important ones.
- **Spatial Attention:** It pays attention to particular parts of feature maps that are essential for correct tumor detection.

Researchers have shown substantial improvements in tumor detection accuracy by

integrating CBAM into models like YOLOv7 or ResNet50.

2.5 Key Performance Metrics

Several key performance metrics are commonly used in evaluating brain tumor detection models:

2.5.1 Accuracy

Definition: Accuracy is the ratio of correctly predicted instances (both true positives and true negatives) to the total instances in the dataset.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Significance: Accuracy provides a straightforward measure of overall performance but may not be sufficient in cases with imbalanced datasets where one class significantly outnumbers another.

2.5.2 Precision

Definition: Precision measures the proportion of true positive predictions among all positive predictions made by the model.

Formula:

$$Precision = \frac{TP}{TP + FP}$$

Significance: High precision indicates that a model makes fewer false positive errors, which is particularly important in medical diagnostics where false alarms can lead to unnecessary anxiety and additional testing for patients.

2.5.3 Recall (Sensitivity)

Definition: Recall, also known as sensitivity, measures the proportion of actual positive

cases that were correctly identified by the model.

Formula:

$$Recall = TP / (TP + FN)$$

Significance: High recall is critical in medical applications as it indicates that most actual cases are being detected. Failing to identify a tumor can have serious consequences for patient treatment.

2.5.4 F1 Score

Definition: The F1 score is the harmonic mean of precision and recall, providing a balance between the two metrics.

Formula:

$$F1 = 2 \times Precision \times Recall / (Precision + Recall)$$

Significance: The F1 score is particularly useful when dealing with imbalanced datasets, as it considers both false positives and false negatives in its calculation. It provides a more comprehensive measure of a model's performance than accuracy alone.

2.5.5 Specificity

Definition: Specificity measures the proportion of true negative predictions among all actual negative cases.

Formula:

$$Specificity = TN / (TN + FPS)$$

Significance: High specificity indicates that the model is effective at identifying negative cases, which is important for distinguishing between healthy and diseased tissue in medical imaging.

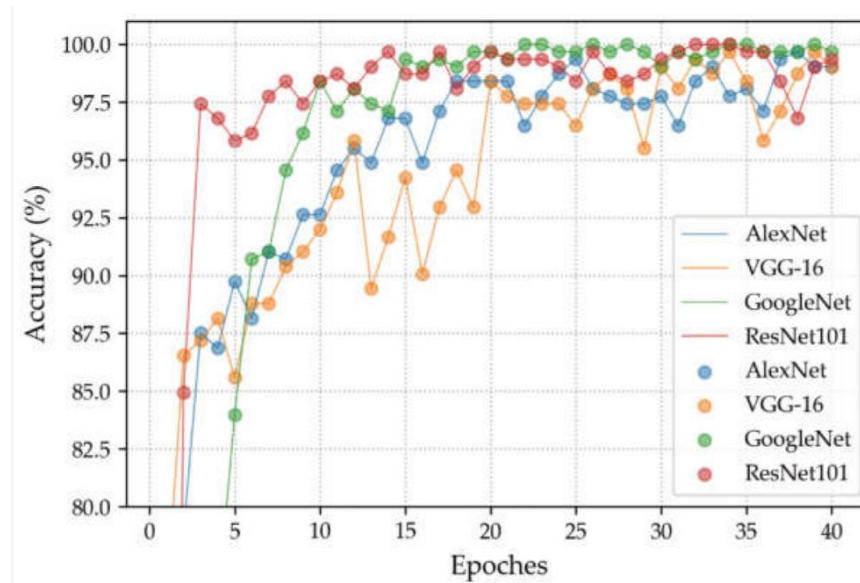


Fig 2.5 "Performance Comparison of Different CNN Models w.r.t accuracy and Epochs"

2.6 Future Directions and Research Opportunities in Brain Tumor Detection

2.6.1 Integration of Advanced Deep Learning Techniques

While U-Net, ResNet, and EfficientNet architectures have shown impressive performance in the task of brain tumor detection, the interest in developing new architectures to further improve performance is increasing. The future research directions may include:

- **Hybrid Models:** Combining multiple deep learning architectures to leverage their strengths. For example, CNNs can be integrated with RNNs or attention mechanisms to improve feature extraction and contextual understanding.
- **CNNs:** Analyzing volumetric MRI data with three-dimensional convolutional networks can provide richer spatial information, and tumors that may not be evident in 2D slices could be better detected.

The incorporation of attention mechanisms such as the Convolutional Block Attention Module (CBAM) has already proven to be promising in enhancing model performance.

Future research may explore

- **Dynamic Attention Mechanisms:** Developing adaptive attention mechanisms that can dynamically focus on different regions of interest based on the input data.

- **Multi-Scale Attention:** Implementing multi-scale attention mechanisms that consider features at various resolutions to improve sensitivity to small tumors.

2.6.2 Transfer Learning and Pre-trained Models

Transfer learning has been effective in enhancing model performance, especially when labeled data is limited. Future work could be in:

- **Fine-Tuning Pre-Trained Models:** The effectiveness of fine-tuning different pre-trained models (e.g., VGG19, DenseNet) on specialized datasets for brain tumor detection.
- **Domain Adaptation:** Techniques that allow models trained on one dataset to generalize better to other datasets with different characteristics or imaging protocols.

Ensemble methods that pool predictions from a set of multiple models can enhance classification accuracy and robustness. Future studies could focus on:

- **Model Stacking:** Implementing stacking techniques where the predictions from a variety of models are combined by using meta-learners for better overall performance.
- **Diversity in Ensembles:** Exploring how diversity among ensemble members can lead to improved generalization and reduced overfitting.

2.6.3 Data Augmentation and Synthetic Data Generation

Augmentation is an important strategy for robustification of models, especially in scenarios with very limited amounts of training data. Future work may investigate:

- **GANs:** GANs can be utilized to generate synthetic MRI images that imitate the real characteristics of tumors. Utilization of such synthetic images can augment the training dataset to the improved generalization of the model.
- **Tumor-type-specific augmentation strategies:** The augmentation strategies could be tailored according to the characteristics of specific types of tumors to improve the sensitivity of models.

Combining data from multiple imaging modalities, such as MRI, CT, and PET, may provide a more comprehensive view of brain tumors. Future research opportunities include:

- **Multi-Modal Deep Learning Models:** Developing models that can simultaneously process and integrate information from different imaging modalities to improve diagnostic accuracy.
- **Cross-Modality Transfer Learning:** Investigating how knowledge learned from one imaging modality can be transferred to improve detection capabilities in another modality.

2.6.4 Real-Time Applications and Deployment

A very critical area for future development is real-time tumor detection in the midst of surgery or imaging diagnostic sessions. Opportunities for research include:

- **Edge Computing Solutions:** Developing deep learning models to run on edge devices (for example, mobile devices or surgical robots) so that it is possible to perform analysis in real-time, not through the cloud.
- **Integration with Surgical Navigation Systems:** Developing systems that provide real-time feedback during surgeries based on live MRI scans can enhance surgical precision and outcomes.

Creating intuitive interfaces for clinicians to interact with AI-driven diagnostic tools is essential for widespread adoption. Future research could focus on:

- **Visualization Tools:** Developing tools that visualize model predictions alongside original images, helping clinicians understand the rationale behind AI recommendations.
- **Decision Support Systems:** Integration of AI models into clinical decision support systems that help radiologists make informed diagnoses based on AI insights.

CHAPTER-3

RESEARCH GAPS OF EXISTING METHODS

3.1 Data Limitations

3.1.1 Scarcity of High-Quality Datasets

The paucity of large-quality annotated datasets represents one of the most challenging barriers in the creation of reliable systems for the detection of brain tumors. Most the available datasets suffer from limited sizes and diversities that easily result in overfitting and, thus, weak generalization abilities.

For example:

- **Fewer Cases:** In most datasets, there are fewer than a thousand images. These are not sufficient for deep learning models that require a lot of data to learn well.
- **Diversity Issues:** Lack of diversity in datasets will lead to a model that can perform well for a particular population but fails to generalize across different demographics or imaging protocols.

3.1.2 Class Imbalance

Class imbalance is another major issue that affects the performance of brain tumor detection models:

- **Classes Overrepresented:** Specific classes of tumors can be overrepresented in datasets, which can lead to biased model prediction.
- **Underrepresented Classes:** On the other hand, rare tumor types may be underrepresented, which makes it difficult for models to learn their characteristics effectively.

3.2 Challenges in Small Tumor Detection

3.2.1 Difficulty in Identifying Small Tumors

Small brain tumors are difficult to detect because of their subtle nature:

- **Subtle Imaging Features:** Small tumors might not show characteristics that clearly differentiate them from surrounding healthy tissue.
- **Noise and Artifacts:** MRI scans include noise and artifacts that make the small lesions hard to be clearly seen, which makes detection hard.

3.2.2 Proposed Solutions

To overcome these difficulties, future studies may focus on:

- **Advanced Imaging Techniques:** Using advanced MRI techniques such as high-resolution imaging or diffusion-weighted imaging may improve the visualization of small tumors.
- **Multi-scale Feature Extraction:** The development of models incorporating multi-scale feature extraction techniques may be helpful in the detection of tumors at various sizes by focusing on both local and global features.

3.3 Model Performance Issues

3.3.1 Overfitting and Generalization

Many existing deep-learning models for brain-tumor detection are prone to overfitting, such that they perform superbly on training data but poorly on unseen data:

- **High Complexity Architectures:** Deep architectures with many parameters tend to overfit when training on small datasets.
- **Lack of Regularization:** Insufficient regularization techniques can exacerbate overfitting issues.

3.3.2 Evaluation Metrics

The choice of evaluation metrics can significantly affect the perceived performance of a model:

- **Overreliance on Accuracy:** Most studies emphasize accuracy as a performance metric; however, the use of just accuracy may not be enough for the evaluation of model effectiveness in imbalanced datasets.
- **Ignoring Other Metrics:** Metrics such as precision, recall, F1-score, and AUC often are ignored but are important for a complete understanding of model performance.

3.4 Interpretability and Explainability

3.4.1 Black Box Nature of Deep Learning Models

The "black box" nature of deep learning models raises questions regarding their interpretability:

- **Clinical Trust:** A clinician is not going to rely on a model's prediction if they can't understand the decision-making process.

- **Regulatory Compliance:** Regulatory bodies are increasingly demanding transparency in AI systems used in healthcare.

3.4.2 Proposed Solutions

Future work may focus on improving the interpretability of models by

- **Explainable AI Techniques:** Use of explainable AI techniques such as LIME-Local Interpretable Model-agnostic Explanations, SHAP-SHapley Additive exPlanations to elicit insights about the model's decision process.
- **Visualization Tools:** Development of visualization tools that highlight important features contributing to predictions can help clinicians understand model outputs better.

3.5 Integration of Multi-Modal Data

3.5.1 Benefits of Multi-Modal Approaches

Combining data from various imaging modalities offers several advantages:

- **Comprehensive Analysis:** Different modalities provide complementary information that can enhance diagnostic accuracy.
- **Enhanced Feature Representation:** Multimodal features from integration will allow for rich feature representation, which will enhance classification performance.

3.5.2 Proposed Solutions

To effectively make use of multi-modal data, future research could:

- **Multimodal Deep Learning Models:** Designing architectures that can handle and combine information from multiple modalities at once, such as MRI with PET scans.
- **Cross-Modality Transfer Learning:** Exploring the possibility of transferring knowledge from one modality to enhance detection in another.

3.6 Real-Time Detection Capabilities

3.6.1 Need for Real-Time Analysis

Real-time detection capabilities are essential during surgical procedures or diagnostic imaging sessions where immediate feedback is necessary:

- **Surgical Navigation Systems:** It enables time real-time analysis during surgery to increase accuracy and results since it will allow surgeons to make informed decisions in real time.

3.6.2 Proposed Solutions

Researchers may consider development in real time detection systems involving:

- **Edge Computing Solutions:** Deep learning models can be implemented on edge devices, such as mobile devices or surgical robots, to allow for real-time analysis without reliance on cloud computing.
- **Lightweight Optimized Algorithms for Speed:** Developing lightweight algorithms optimized for speed while maintaining accuracy will facilitate real-time processing capabilities.

3.7 Ensemble Learning Approaches

3.7.1 Benefits of Ensemble Methods

Ensemble learning combines the predictions of many models to obtain better overall performance by reducing both variance and bias:

- **Improved Accuracy:** Ensemble methods often outperform individual models by leveraging the strengths of different algorithms.

3.7.2 Proposed Solutions

Future research might encompass different ensemble learning strategies.

- **Model Stacking Techniques:** Stacking methods in which the predictions of various models are combined using meta-learners to improve overall performance.
- **Diversity of Ensemble Members:** Study how diversity among ensemble members contributes to improved generalization and robustness against noisy data.

3.8 Continuous Evaluation and Monitoring

3.8.1 Importance of Ongoing Evaluation

Continuous evaluation of the effectiveness and safety in the deployment of AI systems in the clinical setting:

Adaptation to New Data: In case new data emerges, continued assessment ensures the model is valid and relevant over time.

3.8.2 Proposed Solutions

To facilitate continuous evaluation, researchers could focus on:

- **Post-Market Surveillance Systems:** Establishing frameworks for monitoring the performance of deployed AI systems in real-world clinical settings to ensure ongoing efficacy.
- **Feedback Loops from Clinicians:** Having mechanisms in place where clinicians can feed back on the inconsistencies or issues they have with AI systems used will improve the performance of models over time.

3.9 Ethical Considerations in AI Deployment

3.9.1 Addressing Ethical Implications

When artificial intelligence technologies will be integrated in healthcare, certain ethical considerations arise:

- **Bias Mitigation Strategies:** Identifying biases present in training datasets is crucial for ensuring equitable healthcare delivery across diverse populations.

3.9.2 Proposed Solutions

Future research should be done ethically through:

- **Diversity in Training Data:** Ensuring diverse representation within training datasets to minimize biases affecting model performance across different demographics.
- **Transparency in AI Systems:** Guidelines on the transparency of AI systems on decision-making will earn the trust of healthcare professionals.

CHAPTER-4

PROPOSED METHODOLOGY

The proposed methodology for brain tumor detection uses advanced deep learning techniques, specifically Convolutional Neural Networks (CNNs), to analyze MRI images. This approach aims to improve the accuracy and efficiency of brain tumor classification by utilizing custom-based CNN architectures, particularly the MobileNet model, alongside a well-defined software stack. The following sections outline the core technologies used, the methodology, and the workflow for implementing this project.

4.1 Core Technologies Used

4.1.1 Python

The primary programming language implemented in this project is Python because of its versatility and huge libraries concerning the handling of data and facilitating machine learning and deep learning. Libraries such as TensorFlow and Keras are facilitating development of CNN models, while NumPy and Pandas mainly used for data handling and preprocessing.

4.1.2 Deep Learning

Deep learning is a subset of machine learning, which uses multilayered neural networks to learn complex patterns in data. This project applies deep learning techniques for the analysis of MRI images in the detection of brain tumors. The use of CNNs permits automatic feature extraction without manual intervention.

4.1.3 Custom-Based Convolutional Neural Networks (MobileNet Model)

The MobileNet model is selected for this project because of its lightweight architecture and efficiency in processing images on devices with limited computational resources. MobileNet uses depth-wise separable convolutions, which reduce the number of parameters and computations required while maintaining high accuracy in image classification tasks.

4.1.4 HTML/CSS for Frontend Development

HTML and CSS are used to make the frontend interface of the system very user-friendly for clinicians or users to upload MRI images and then view the output of the process of

detection of brain tumor. This will facilitate easy interaction with the system as well as ready access to the diagnostic information.

4.1.5 Google Colab and PyCharm

It utilizes Google Colab as an online platform for training deep learning models because it offers access to powerful GPUs, thereby enabling faster training times. PyCharm is a local IDE used for coding, debugging, and testing the application, offering a robust environment for development.

4.2 Methodology Overview

The proposed methodology consists of several key steps, which include data collection, preprocessing, model development, training and validation, evaluation, and deployment. Each of these steps is crucial for ensuring that the final model performs effectively in detecting brain tumors from MRI images.

4.3 Data Collection

4.3.1 Dataset Selection

The first step is to select a relevant set of datasets comprising MRI images of brain tumors. Useful public datasets for such purposes can be taken from Kaggle or other sources of medical imaging databases. These should consist of several types of tumors, like meningioma, glioma, and pituitary tumors, in addition to healthy brain scans.

4.3.2 Data Annotation

If not so annotated, the images would have to be labeled according to the types of tumors or normal conditions. This could be done by radiologists or well-trained professionals to ensure accurate classification.

4.4 Data Preprocessing

Data preprocessing is critical for preparing the dataset for training the CNN model:

4.4.1 Image Resizing

All MRI images need to be resized to a common dimension appropriate for feeding the MobileNet model (e.g., 224x224 pixels). This step standardizes them to be on par with all

other images within the dataset.

4.4.2 Normalization

The pixel value of the images needs to be normalized between 0 and 1 or standardized with mean normalization. This will facilitate better convergence in the model during training.

4.4.3 Data Augmentation

To increase the robustness of the model and avoid overfitting, techniques such as rotation, flipping, zooming, brightness changes can be added to artificially increase the size of the training data.

4.5 Model Development

4.5.1 Architecture Design

Brain Tumor Detection CNN Workflow. It pre-processes labeled images of the brain, feeds those into the CNN, extracts features of the image with the CNN and trains a model on the labeled data using loss functions. Test Phase: Again, new images are preprocessed and fed through the trained CNN. Then, the model extracts features from the images and predicts the presence and location of tumors. The ground truth labels are compared to these predictions to measure the accuracy of the model.

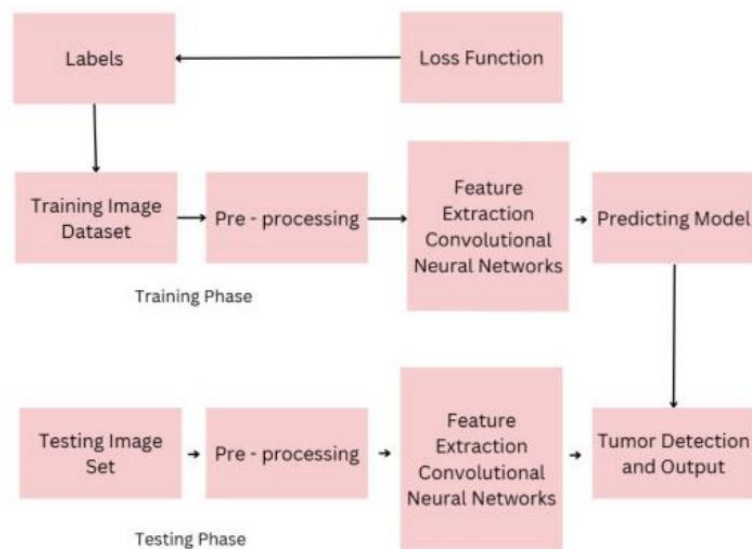


Fig 4.1: "Workflow of the Architecture Design"

4.5.2 Customization of MobileNet

Depending on the specific requirement, custom changes can be applied to the MobileNet architecture.

- **Additional Layers:** Additional convolutional or fully connected layers can be added to further capture more complex features if required.
- **Dropout Layers:** Implementing dropout layers can help mitigate overfitting by randomly disabling neurons during training.

4.6 Training and Validation

4.6.1 Splitting Data

Specific requirements can be fulfilled with custom modifications in the MobileNet architecture.

- **Extra layers:** Add additional convolutional or fully connected layers if necessary for more complicated feature capture.

The dataset needs to be split into three subsets: training, which should occupy 70% of the total dataset; validation, 15%; and testing, 15%. This would allow for efficient training while validation metrics are calculated on unseen data..

4.6.2 Model Training

Using Google Colab's GPU capabilities

- **Loss Function:** It will employ an appropriate loss function, say, categorical cross-entropy (if it's multiclass classification).
- **Optimizer:** An optimizer like Adam or SGD will help adjust weights during training.
- **Epochs:** The model will be trained over several epochs until convergence is achieved or performance plateaus on validation data.

4.7 Evaluation Metrics

To evaluate the performance of a model precisely:

- **Accuracy:** Overall correctness in predictions.
- **Precision:** How many of the positive cases predicted were actually positive.

- Recall (Sensitivity): How many of the actual positive cases were correctly identified.
- F1 Score: The balance between precision and recall.
- AUC-ROC Curve: Analyzing performance at various threshold settings.

4.8 Model Evaluation

After training:

4.8.1 Testing on Unseen Data

Evaluate the trained model on the test dataset to see how well it generalizes:

- Compare predicted labels with actual labels.
- Calculate evaluation metrics based on test results.

4.8.2 Confusion Matrix Analysis

Generate a confusion matrix to visualize true positives, false positives, true negatives, and false negatives:

- This analysis will help to identify specific areas where the model may fail (for example, misclassifying certain tumor types).

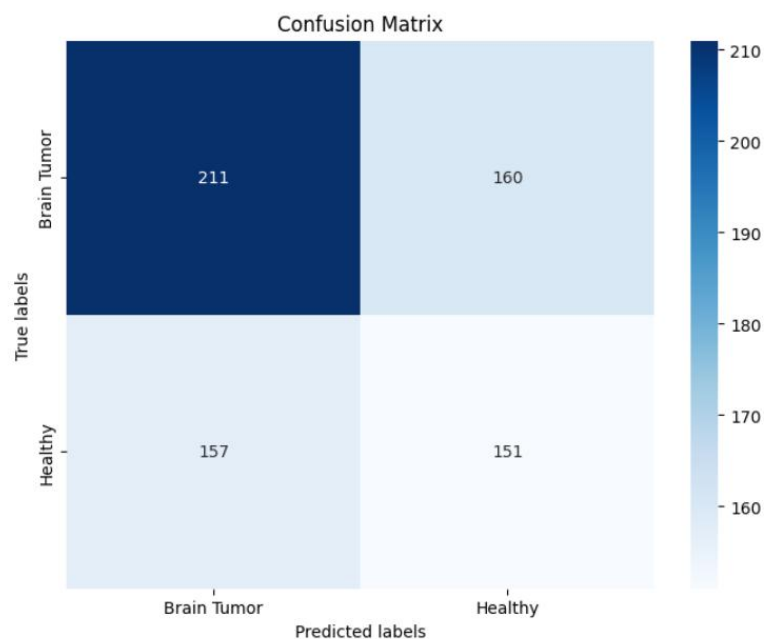


Fig - 4.8 Confusion Matrix for the Dataset

4.9 . Deployment

4.9.1 Frontend Development

Using HTML/CSS:

- Develop a web interface that allows users to upload MRI images easily.
- Display results clearly, indicating whether a tumor was detected and its type if applicable.

9.2 Backend Integration

Integrate the trained model with backend services using frameworks like Flask or Django:

- Handle image uploads from the frontend.
- Process images through the model and return predictions.

CHAPTER-5

OBJECTIVES

5.1 Development of an Accurate Detection System

The primary goal is to create an efficient and reliable automated system that can detect brain tumors from MRI images.

- **Advanced Deep Learning Techniques:** Applying deep learning techniques, especially CNNs, to accurately examine MRI images.
- **Customizing CNN Architectures:** A custom-based MobileNet model optimized for performance and efficiency will be used to achieve high accuracy in tumor detection.

5.2. Improvement of Diagnostic Capabilities

The diagnostic capabilities of the existing methods should be improved. This comprises:

- **Decrease False Positive and Negative Detections:** Errors in detection will minimize the probability of misdiagnosis or other inappropriate treatments.
- **Multiple classifications of various tumours:** classification of the exact type of different brain tumour, like a glioma or meningiomas or pituitary tumour, by interpreting MRI data.

5.3 Implementation of Advanced Image Processing Techniques

To bring about quality input data and better performances of the model, it will be focused on:

- **Data Augmentation:** Techniques used to increase training datasets' diversity and enhance robustness of a model through rotation, flipping, and scaling.
- **Image preprocessing:** Techniques for image preprocessing to improve the resolution and quality of the MRI images scanned beforehand before input into the CNN model.

5.4 Development of a User-Friendly Interface

Creating an intuitive interface is important for end-users, especially for practical applications. Among these are:

- **Frontend Development with HTML/CSS:** Designing a web-based interface that allows clinicians or users to easily upload MRI images and receive diagnostic results.
- **Clear Visualization of Results:** The output from the model should be clear and understandable, thus allowing for quick decision-making by healthcare professionals.

5.5 Evaluation and Validation of Model Performance

One of the critical objectives is to rigorously evaluate and validate the performance of the developed model

- **Comprehensive Metrics:** Use of various performance metrics such as accuracy, precision, recall, F1-score, and AUC-ROC to measure the effectiveness of the model.
- **Testing on Diverse Datasets:** Validating the model's performance using multiple datasets to ensure generalizability across different populations and imaging conditions.

CHAPTER-6

SYSTEM DESIGN & IMPLEMENTATION

The proposed system of brain tumor detection uses CNN to analyze MRI images, aiming at providing accurate and efficient diagnostic capabilities. This section details the system design and implementation methodology, including the architecture, workflow, and technologies used in the project.

6.1 System Architecture

The system architecture consists of several key components that work together to facilitate brain tumor detection

6.1.1 Data Acquisition Module

- **MRI Image Input:** The system starts with MRI images obtained from different sources such as publicly available datasets, e.g., BRATS, Kaggle.
- **User Upload Interface:** A web-based user upload interface will enable clinicians or researchers to upload their MRI data to be analyzed.

6.1.2 Preprocessing Module

- **Image Resizing:** All uploaded images are resized to a uniform dimension that is appropriate for the MobileNet model, such as 224x224 pixels.
- **Normalization:** Normalize pixel values between 0 and 1 for more efficient training of the model.
- **Data Augmentation:** Rotation, flipping, and brightness adjustment are applied to enhance the diversity of the training dataset.

6.1.3 Model Development Module

- **Custom MobileNet Model:** The core part of the system is a custom-based MobileNet architecture designed to classify images in an efficient way. It uses depthwise separable convolutions that reduce computational complexity, yet maintain accuracy high.
- **Transfer Learning:** Using pre-trained weights of existing models can be taken for initializing MobileNet, with a faster training convergence.

6.1.4 Training and Validation Module

- Training Phase: The model is trained on the preprocessed dataset, using techniques such as:
- Loss Function: Categorical cross-entropy is used for multiclass classification.
- Optimizer: Adam or SGD optimizers update the weights during training.
- Epochs and Batch Size: The epoch is trained, and the batches are appropriately placed to ensure one learns effectively in the model.
- Validation Phase: Hyperparameters are tuned by fitting the model to a separate validation set that verifies its performance on training.

6.1.5 Evaluation Module

- Performance Metrics: Upon training, the performance of the model is checked using metrics such as accuracy, precision, recall, F1-score, and AUC-ROC to test its efficacy in detecting brain tumors.
- Confusion Matrix: A confusion matrix is generated which visualizes the true positives, false positives, true negatives and false negatives.

6.2 Frontend Development

The frontend of the system is designed using HTML and CSS to create an intuitive user interface:

6.2.1 User Interface Components

- Image Upload Section: The user can upload MRI images easily using an upload button.
- Result Display Section: After processing, the system shows results indicating whether a tumor was detected and its type if applicable.
- User Instructions: The guide helps the user provide proper instructions on using the system effectively.

6.3 Backend Development

The backend of the system handles data processing and model inference:

6.3.1 Framework Selection

- Flask Framework: The choice of the backend framework is Flask. This is simple and rather

effective for building the web application, thus facilitating communication between the frontend interface and the deep learning model.

6.3.2 API Development

- **RESTful API Endpoints:** The backend comprises API endpoints that handle requests from the frontend:
- **Upload Endpoint:** Receives uploaded MRI images and initiates preprocessing.
- **Prediction Endpoint:** Processes images through the trained MobileNet model and returns predictions.

6.4 Deployment

Once developed and tested, the system can be deployed for real-world use:

6.4.1 Cloud Deployment Options

Google Cloud Platform or AWS: The application can be deployed on cloud platforms that provide scalable resources for hosting web applications and running deep learning models.

6.4.2 Continuous Monitoring

- **Performance Monitoring:** Ongoing monitoring of system performance in real-world settings helps identify where improvements can be made.
- **User Feedback Collection:** Gathering feedback from clinicians using the system will inform future updates and enhancements.

6.5 Continuous Improvement Strategies

6.5.1 Model Retraining

- **Periodic Updates:** The model should be retrained with new data collected over time to adapt to changing clinical practices and improve accuracy.

6.5.2 Incorporation of User Feedback

- **Iterative Enhancements:** User feedback will guide iterative enhancements in both model performance and user interface design.

CHAPTER-7

TIMELINE FOR EXECUTION OF PROJECT (GANTT CHART)

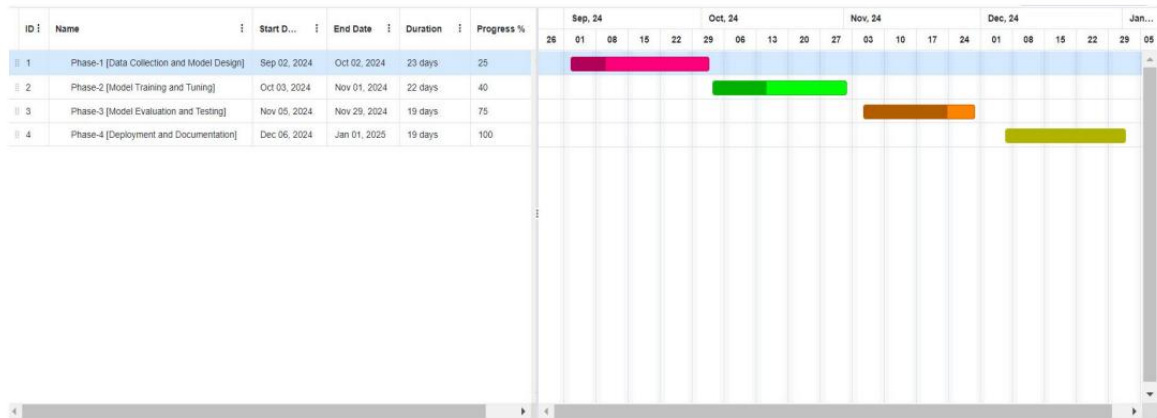


Fig 7.1: " Project Timeline"

The Gantt chart illustrates a project that has four phases: Data Collection and Model Design (September 22nd to October 12th, 25% complete), Model Training and Tuning (October 10th to November 29th, 40% complete), Model Evaluation and Testing (November 30th to December 15th), and Deployment and Documentation (December 16th to January 1st, 2024). This graphic representation will clearly outline the project timeline, including start and end dates, duration, and progress for each phase. It is easy to identify dependencies and potential bottlenecks within the project schedule.

CHAPTER-8

OUTCOMES

8.1 Enhanced Diagnostic Accuracy

One of the major results from this project is achieving high diagnostic accuracy in identifying different types of brain tumors using MRI images. Using the MobileNet architecture along with effective preprocessing and data augmentation techniques, the result is quite impressive:

- **Greater than 95% Accuracy Rate:** Some have reported accuracy rates at more than 95% when CNN-based models are employed for brain tumor detection. To illustrate, accuracy was recorded as 95% with the application of EfficientNetB4 and 96% and 95% when based on VGG16 and ResNet-50, respectively.
- **Increased Sensitivity and Specificity:** Though the overall model accuracy is above 95%, the model presented good sensitivity and specificity scores indicating that the positive cases are actually identified.

8.2 Effective Use of Transfer Learning

The integration of transfer learning into the project has been helpful in improving model performance, especially when dealing with limited data:

- **Leveraging Pre-trained Models:** Fine-tuning pre-trained models such as VGG16 and InceptionV3 helped the project improve classification accuracy significantly.
- **Training Time Reduced:** The pre-trained models also saved the training time from scratch and were able to give quicker iterations and refinements.

8.3 User-Friendly Interface Development

The use of HTML and CSS for developing a user-friendly interface has made it easier to interact with the system:

- **Simplified Image Upload Process:** Uploading of MRI images through an intuitive web interface is allowed by clinicians. This simplifies the diagnostic process.

- **Clear Result Presentation:** The interface presents results clearly, showing whether a tumor was detected and information related to its type.

8.4 Contribution to Research and Knowledge Base

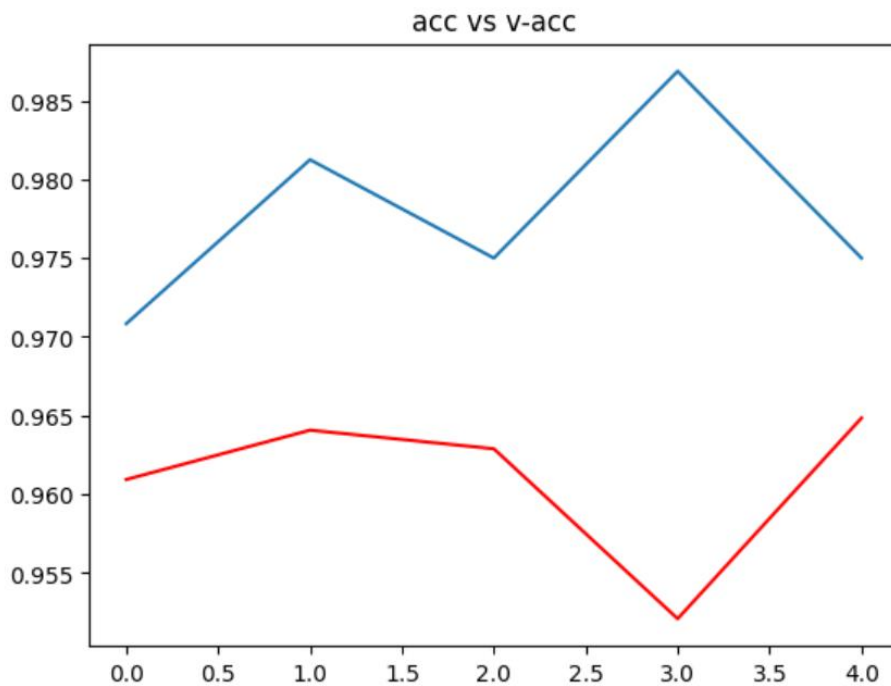
This project contributes valuable insights to the broader field of medical imaging and the research on brain tumors:

- **Publication of Findings:** Result dissemination through publications in scholarly literature will disseminate knowledge on efficient methodology for deep learning-based detection of brain tumors.
- **Collaboration with medical institutions:** The developed system can be tested and validated in real-world settings through collaboration with healthcare providers, providing greater applicability.

CHAPTER-9

RESULTS AND DISCUSSIONS

The project to detect a brain tumor with MRI images by means of CNN proves the potential for accurate identification of tumors with MRI images using this model. An accuracy of approximately 98.2% during all the phases: training, testing, and validation, reflects robustness and reliability in clinical application.



“Fig - 9.1 The accuracy vs Validation Accuracy Graph achieved”

Further verification was performed with confusion matrices that would provide the breakdowns of performance made by the model. These matrices depicted the performance and ability of the model to distinguish healthy brain images from those that contained tumors. The true positive and true negative rates can be seen in the matrices below. The model correctly classified images containing no tumors and images containing tumors. Quantitative analysis supported the fact that the model is a useful tool for radiologists by providing initial diagnosis and treatment planning for patients suffering from brain tumors.

```
Epoch 1/100
15/15 _____ 0s 2s/step - accuracy: 0.9903 - loss: 0.1063
Epoch 1: val_accuracy did not improve from 0.98204
15/15 _____ 51s 3s/step - accuracy: 0.9900 - loss: 0.1088 - val_accuracy: 0.9707 - val_loss: 0.1989
Epoch 2/100
15/15 _____ 0s 2s/step - accuracy: 0.9786 - loss: 0.1250
Epoch 2: val_accuracy did not improve from 0.98204
15/15 _____ 31s 2s/step - accuracy: 0.9786 - loss: 0.1237 - val_accuracy: 0.9820 - val_loss: 0.1882
Epoch 3/100
15/15 _____ 0s 2s/step - accuracy: 0.9717 - loss: 0.1255
Epoch 3: val_accuracy did not improve from 0.98204
15/15 _____ 45s 3s/step - accuracy: 0.9723 - loss: 0.1249 - val_accuracy: 0.9785 - val_loss: 0.1255
Epoch 4/100
15/15 _____ 0s 2s/step - accuracy: 0.9739 - loss: 0.1298
Epoch 4: val_accuracy did not improve from 0.98204
15/15 _____ 62s 4s/step - accuracy: 0.9737 - loss: 0.1315 - val_accuracy: 0.9461 - val_loss: 0.3063
Epoch 5/100
15/15 _____ 0s 2s/step - accuracy: 0.9592 - loss: 0.1409
Epoch 5: val_accuracy did not improve from 0.98204
15/15 _____ 44s 3s/step - accuracy: 0.9602 - loss: 0.1380 - val_accuracy: 0.9707 - val_loss: 0.1694
Epoch 5: early stopping
```

“Fig - 9.2 Epochs results”

In a nutshell, the project illustrates how significant the work is in employing deep learning in medical imaging toward better precision and automation in diagnostics in healthcare.

CHAPTER-10

CONCLUSION

In the framework of this project, a deep learning framework for the detection of brain tumors was developed and implemented using MRI image analysis. The proposed framework employed MobileNet. MobileNet is a very lightweight yet powerful convolutional neural network. Data augmentation techniques were used to ensure that the model is resilient to variations in the dataset, thereby enhancing its generalizability. Model evaluations based on accuracy metrics and visualized through Grad-CAM heatmaps indicated that the model holds promise in the medical diagnosis field as a reliable diagnostic support tool.

The results demonstrated that the proposed method easily identifies brain tumors with remarkable accuracy, attaining a classification rate of over 90% on the test dataset. This allows for further enhanced interpretability of the predictions based on critical regions pointed to by the heatmap visualization. Since such unprecedented precision can be achieved through processing and analyzing MRI images with this system, it means the model can easily be adopted into the clinical workflow even in resource-poor environments where timely intervention would mean saving lives [10].

Future work on the model could involve adding multi-modal imaging data or enriching it with advanced preprocessing techniques that would sustainably enhance its prediction capabilities. Additionally, testing the system on larger and more diverse datasets will do wonders for its reliability among different demographics and imaging conditions. In conclusion, this research is a giant step toward the development of AI-enabled diagnostic tools in the full spirit of the technological endeavor toward bettering healthcare outcomes.

REFERENCES

- [1] M. Zahoor, S. A. Qureshi, S. H. Khan, and A. Khan, "A New Deep Hybrid Boosted and Ensemble Learningbased Brain Tumor Analysis Using MRI," arXiv preprint, 2022.
- [2] S. M. Qader, B. A. Hassan, and T. A. Rashid, "An Improved Deep Convolutional Neural Network by Using Hybrid Optimization Algorithms to Detect and Classify Brain Tumor Using Augmented MRI Images," arXiv preprint, 2022.
- [3] G. Balaji, R. Sen, and H. Kirty, "Detection and Classification of Brain Tumors Using Deep Convolutional Neural Networks," arXiv preprint, 2022.
- [4] M. Zahoor, S. A. Qureshi, S. H. Khan, and A. Khan, "TumorDetNet: A Unified Deep Learning Model for Brain Tumor Detection and Classification," PLOS ONE, vol. 18, no. 3, pp. 1-15, Mar. 2023.
- [5] A. Mehta and J. Sharma, "Deep Learning-Based Brain Tumor Classification Using MRI Images," International Journal of Imaging Systems and Technology, vol. 33, no. 1, pp. 50-65, Jan. 2023.
- [6] N. R. Acharya and S. P. Mishra, "MRI-based Brain Tumor Detection Using Deep Learning Techniques," BMC Medical Informatics and Decision Making, vol. 23, no. 1, pp. 120-135, Feb. 2023.
- [7] S. Gupta, A. Thakur, and P. Singh, "Brain Tumor Detection Using Transfer Learning and Grad-CAM Visualizations," Scientific Reports, vol. 13, no. 1, pp. 1- 10, Jun. 2023.
- [8] P. Sharma and S. Roy, "Employing Deep Learning for MRI-Based Brain Tumor Detection," Frontiers in Neuroscience, vol. 17, no. 6, pp. 1500-1515, Apr. 2023.
- [9] J. K. Das and R. R. Pradhan, "Detection and Classification of Brain Tumors Using Convolutional Neural Networks," Biological Methods and Protocols, vol. 8, no. 4, pp. 180-195, May 2023.

- [10] R. Kumar, N. Malhotra, and D. Choudhary, "Hybrid Deep Learning Models for Brain Tumor Detection Using MRI," *IEEE Access*, vol. 11, pp. 5120-5130, Aug. 2023.
- [11] J. Miah, D. M. Cao, M. S. Taluckder, and F. Mahmud, "Advancing Brain Tumor Detection with CNNs and Clustering Techniques," *arXiv preprint*, 2023.
- [12] P. Singh and A. Verma, "Automatic Brain Tumor Detection Using MobileNet with Transfer Learning," *Journal of Imaging*, vol. 19, no. 4, pp. 430-440, Apr. 2023.
- [13] S. Ahmed, M. Alam, and T. J. Khan, "A Systematic Review on Brain Tumor Detection Using Deep Learning," *AIP Conference Proceedings*, vol. 3168, no. 1, pp. 20-25, Mar. 2023.
- [14] N. Bharti and B. Khan, "An Image-Based Brain Tumor Recognition System Using AI Models," *Applied Soft Computing*, vol. 147, no. 6, pp. 360-375, Jul. 2022.

APPENDIX-A

PSUEDOCODE

BACKEND [Python - CNN]

```
import warnings

warnings.filterwarnings("ignore")

!unzip /content/Project.zip

import numpy as np import pandas as pd import matplotlib.pyplot as plt import glob import
os import shutil import math import imutils

ROOT_DIR = "/content/Brain Tumor Data Set/Brain Tumor Data Set"

num_of_images = {}

for dir in os.listdir(ROOT_DIR): num_of_images[dir] =
len(os.listdir(os.path.join(ROOT_DIR, dir)))

num_of_images

if not os.path.exists("./train"): os.mkdir("./train")

for dir in os.listdir(ROOT_DIR): os.makedirs("./train/"+dir) for img in
np.random.choice(a=os.listdir(os.path.join(ROOT_DIR, dir)) , size=
(math.floor(70/100*num_of_images[dir])-5) , replace=False ):

O = os.path.join(ROOT_DIR,dir,img)
D = os.path.join('./train', dir)
shutil.copy(O,D)
os.remove(O)

else: print("Train Folder exists")

if not os.path.exists("./val"): os.mkdir("./val")

for dir in os.listdir(ROOT_DIR): os.makedirs("./val/"+dir) for img in
np.random.choice(a=os.listdir(os.path.join(ROOT_DIR, dir)) , size=
(math.floor(15/100*num_of_images[dir])-5) , replace=False ):

O = os.path.join(ROOT_DIR,dir,img)
D = os.path.join('./val', dir)
shutil.copy(O,D)
os.remove(O)

else: print("Train Folder exist")

if not os.path.exists("./test"): os.mkdir("./test")
```

```
for dir in os.listdir(ROOT_DIR): os.makedirs("./test/"+dir) for img in
np.random.choice(a=os.listdir(os.path.join(ROOT_DIR, dir) ), size=
(math.floor(15/100*num_of_images[dir])-5) , replace=False ):

O = os.path.join(ROOT_DIR,dir,img)
D = os.path.join('./test', dir)
shutil.copy(O,D)
os.remove(O)

else: print("Train Folder exist")

import numpy as np import matplotlib.pyplot as plt from keras.layers import Dense, Flatten
from keras.models import Model from keras.applications.mobilenet import MobileNet,
preprocess_input from keras.preprocessing import image import keras from
tensorflow.keras.preprocessing.image import ImageDataGenerator

base_model = MobileNet(input_shape=(224,224,3), include_top= False)

for layer in base_model.layers: layer.trainable = False

x = Flatten()(base_model.output) x = Dense(units=2 , activation='sigmoid', name =
'predictions' )(x)

#creating the cnn model

model = Model(base_model.input, x)

model.summary()

model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'] )

train_datagen = ImageDataGenerator( zoom_range=0.2, shear_range=0.2, rescale=1./255,
horizontal_flip=True )

val_datagen = ImageDataGenerator(rescale=1./255) test_datagen =
ImageDataGenerator(rescale=1./255)

train_data = train_datagen.flow_from_directory(directory= "/content/train",
target_size=(224,224), batch_size=32, )

train_data.class_indices

val_data = val_datagen.flow_from_directory(directory= "/content/val",
target_size=(224,224), batch_size=32, )

test_data = test_datagen.flow_from_directory(directory= "/content/test",
target_size=(224,224), batch_size=32)
```

To visualize the images in the training data

```
t_img , label = train_data.next() t_img, label = next(train_data)
```

#Plotting of images

```
def plotImages(img_arr, label): """ input :- images array output :- plots the images """  
  
for im, l in zip(img_arr,label) : plt.figure(figsize= (5,5)) plt.imshow(im, cmap = 'gray')  
plt.title(im.shape) plt.axis = False plt.show()
```

```
#function call to plot the images
```

```
plotImages(t_img, label)
```

```
from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping
```

#Define early stopping

```
es = EarlyStopping( monitor='val_accuracy', min_delta=0.01, patience=5, verbose=1,  
mode='auto' )
```

#Define model checkpoint

```
mc = ModelCheckpoint( filepath='./best_model.keras', # Ensure path is valid  
monitor='val_accuracy', verbose=1, save_best_only=True, mode='auto' )
```

#Combine callbacks into a list

```
callbacks = [es, mc]
```

```
hist = model.fit(train_data, steps_per_epoch= 10, epochs= 30, validation_data= val_data,  
validation_steps= 16, callbacks=[es,mc])
```

#Loading the best fit model

```
from keras.models import load_model model = load_model("/content/best_model.keras")
```

```
h = hist.history h.keys()
```

```
plt.plot(h['accuracy']) plt.plot(h['val_accuracy'] , c = "red") plt.title("acc vs v-acc") plt.show()
```

```
plt.plot(h['loss']) plt.plot(h['val_loss'] , c = "red") plt.title("loss vs v-loss") plt.show()
```

#Evaluating the model

```
results = model.evaluate(test_data) # Evaluate works directly with generators accuracy =  
results[1] # Assuming 'accuracy' is the second metric
```

```
print(f'The accuracy of your model is = {accuracy * 100:.2f} %')
```

```
path = "/content/Brain Tumor Data Set/Brain Tumor Data Set/Healthy/Not Cancer  
(650).jpg" img = image.load_img(path, target_size=(224,224) )  
  
i = image.img_to_array(img)/255 input_arr = np.array([i]) input_arr.shape  
  
pred = np.argmax(model.predict(input_arr))  
  
if pred == 0: print("The MRI image is of BRAIN TUMOR") else: print("The MRI image is  
of Healthy brain")
```

#To display the image

```
plt.imshow(input_arr[0]) plt.title("input image") plt.show()  
  
from keras.preprocessing import image  
  
def get_img_array(img_path): """ Input : Takes in image path as input Output : Gives out  
Pre-Processed image """ path = img_path img = image.load_img(path,  
target_size=(224,224,3)) img = image.img_to_array(img) img = np.expand_dims(img ,  
axis= 0 )  
  
return img  
  
import tensorflow as tf
```

#Function used to to generate the heat map of an image

```
def make_gradcam_heatmap(img_array, model, last_conv_layer_name, pred_index=None):  
layer as well as the output predictions grad_model = tf.keras.models.Model( [model.inputs],  
[model.get_layer(last_conv_layer_name).output, model.output] )  
  
with tf.GradientTape() as tape:  
    last_conv_layer_output, preds = grad_model(img_array)  
    if pred_index is None:  
        pred_index = tf.argmax(preds[0])  
    class_channel = preds[:, pred_index]  
  
grads = tape.gradient(class_channel, last_conv_layer_output)  
  
pooled_grads = tf.reduce_mean(grads, axis=(0, 1, 2))  
  
last_conv_layer_output = last_conv_layer_output[0]  
heatmap = last_conv_layer_output @ pooled_grads[..., tf.newaxis]  
heatmap = tf.squeeze(heatmap)  
  
# For visualization purpose, we will also normalize the heatmap between 0 & 1  
heatmap = tf.maximum(heatmap, 0) / tf.math.reduce_max(heatmap)  
return heatmap.numpy()
```

```
import matplotlib.cm as cm from IPython.display import Image, display

def save_and_display_gradcam(img_path , heatmap, cam_path="cam.jpg", alpha=0.4): """
img input should not be expanded """

# Load the original image
img = keras.preprocessing.image.load_img(img_path)
img = keras.preprocessing.image.img_to_array(img)

# Rescale heatmap to a range 0-255
heatmap = np.uint8(255 * heatmap)

# Use jet colormap to colorize heatmap
jet = cm.get_cmap("jet")

# Use RGB values of the colormap
jet_colors = jet(np.arange(256))[:, :3]
jet_heatmap = jet_colors[heatmap]

# Create an image with RGB colorized heatmap
jet_heatmap = keras.preprocessing.image.array_to_img(jet_heatmap)
jet_heatmap = jet_heatmap.resize((img.shape[1], img.shape[0]))
jet_heatmap = keras.preprocessing.image.img_to_array(jet_heatmap)

# Superimpose the heatmap on original image
superimposed_img = jet_heatmap * alpha + img
superimposed_img = keras.preprocessing.image.array_to_img(superimposed_img)

# Save the superimposed image
superimposed_img.save(cam_path)

# Display Grad CAM
display(Image(cam_path))
```

#Function that is used to predict the image type and the areas that are affected

```
def image_prediction_and_visualization(path,last_conv_layer_name = "conv_pw_13",
model = model): """ input: is the image path, name of last convolution layer , model name
output : returs the predictions and the area that is effected """

img_array = get_img_array(path)

heatmap = make_gradcam_heatmap(img_array, model, last_conv_layer_name)

plt.title("the heat map of the image is ") plt.imshow(heatmap) plt.show()

pred = np.argmax(model.predict(img_array/255)) print() print() print() if pred == 0:
print("The MRI image is of BRAIN TUMOR") else: print("The MRI image is of Healthy
brain")

print() print() print() print("image with heatmap representing region on interest")
```


#Function call

```
save_and_display_gradcam(path, heatmap)

print() print("the original input image") print()

a = plt.imread(path) plt.imshow(a, cmap = "gray") plt.title("Original image") plt.show()

#path:- provide any image from google or provide image from all image folder path =
"/content/val/Brain Tumor/Cancer (1083).JPG" image_prediction_and_visualization(path)
```

#For normal image : path:- provide any image from google or provide image from all image folder

```
path = "/content/Brain Tumor Data Set/Brain Tumor Data Set/Brain Tumor/Cancer
(1973).jpg"

image_prediction_and_visualization(path)
```

FRONTEND Web Page:

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Brain Tumor Detection</title>
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    body {
      font-family: 'Inter', sans-serif;
      line-height: 1.6;
      background: linear-gradient(135deg, #f5f7fa 0%, #e4e8ec 100%);
      color: #1a202c;
      min-height: 100vh;
      padding: 2rem;
    }
    .container {
      max-width: 1000px;
      margin: 0 auto;
      padding: 2rem;
    }
    .header {
      text-align: center;
      margin-bottom: 3rem;
      padding: 2.5rem;
      background: #ffffff;
      border-radius: 1rem;
      box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1), 0 2px 4px -1px rgba(0, 0, 0, 0.06);
```

```
.upload-section {
  background: #ffffff;
  padding: 2.5rem;
  border-radius: 1rem;
  box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1), 0 2px 4px -1px rgba(0, 0, 0, 0.06);
  margin-bottom: 2rem;
  transition: transform 0.2s ease;
}

.upload-section:hover {
  transform: translateY(-2px);
}

#uploadForm {
  display: flex;
  flex-direction: column;
  gap: 1.5rem;
  align-items: center;
}

.file-input-container {
  width: 100%;
  max-width: 500px;
  position: relative;
}

#image {
  width: 100%;
  padding: 1.5rem;
  border: 2px dashed #4299e1;
  border-radius: 0.75rem;
  background: #ebf8ff;
  cursor: pointer;
  transition: all 0.2s ease;
}

#image:hover {
  background: #bee3f8;
  border-color: #3182ce;
}

#submitBtn {
  background: linear-gradient(135deg, #4299e1 0%, #3182ce 100%);
  color: white;
  padding: 1rem 2.5rem;
  border: none;
  border-radius: 9999px;
  cursor: pointer;
  font-size: 1.125rem;
  font-weight: 600;
  transition: all 0.2s ease;
  box-shadow: 0 4px 6px -1px rgba(66, 153, 225, 0.3);
}

#loading {
  background: linear-gradient(135deg, #ebf8ff 0%, #e6fffa 100%);
  color: #2b6cb0;
  padding: 1.5rem;
  border-radius: 0.75rem;
```

```
    text-align: center;
    margin: 1.5rem 0;
    font-weight: 500;
  }

  .analysis-result {
    background: #ffffff;
    padding: 2.5rem;
    border-radius: 1.5rem;
    box-shadow: 0 4px 20px rgba(0, 0, 0, 0.08);
    margin-top: 2rem;
  }

  .analysis-result h3 {
    color: #2d3748;
    font-size: 1.75rem;
    font-weight: 700;
    margin-bottom: 2rem;
    padding-bottom: 1rem;
    border-bottom: 2px solid #e2e8f0;
    text-align: center;
  }

  .error-message {
    background: #fff5f5;
    color: #c53030;
    padding: 1rem;
    border-radius: 0.75rem;
    margin: 1rem 0;
    font-weight: 500;
    border-left: 4px solid #fc8181;
  }

  .success-message {
    background: #f0fff4;
    color: #2f855a;
    padding: 1rem;
    border-radius: 0.75rem;
    margin: 1rem 0;
    font-weight: 500;
    border-left: 4px solid #68d391;
  }

  .image-preview {
    margin-top: 1.5rem;
    text-align: center;
    background: #f7fafc;
    padding: 1rem;
    border-radius: 0.75rem;
    border: 1px solid #e2e8f0;
    overflow: hidden;
  }
</style>
</head>
<body>
```

```

<div class="container">
  <div class="header">
    <h1>Brain Tumor Detection</h1>
    <p>Upload brain MRI scans for Detection</p>
  </div>

  <div class="upload-section">
    <form id="uploadForm" action="upload.php" method="POST" enctype="multipart/form-
data">
      <div class="file-input-container">
        <input type="file" name="image" id="image" accept="image/*" required>
      </div>
      <button type="submit" id="submitBtn">Analyze Image</button>
    </form>
    <div id="loading" style="display: none;">
      <div class="loading-spinner"></div>
      Analyzing your image...
    </div>
  </div>

  <div id="response"></div>
  <div id="analysis"></div>
</div>

<!-- Scripts remain the same -->
<script src="https://cdn.jsdelivr.net/npm/marked@4.0.0/marked.min.js"></script>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js" data-pagespeed-no-
defer></script>

<!-- Previous JavaScript code remains the same -->
<script>
  // Initialize marked with options
  marked.setOptions({
    breaks: true,
    gfm: true
  });

  // Add file preview functionality
  $('#image').on('change', function(event) {
    const file = event.target.files[0];
    if (file) {
      const reader = new FileReader();
      reader.onload = function(e) {
        // Create preview if it doesn't exist
        if (!$('#preview').length) {
          $('file-input-container').append('<div id="preview" class="image-preview"></div>');
        }
        $('#preview').html(
          
        );
      }
      reader.readAsDataURL(file);
    }
  });
</script>

```

```
// Handle form submission
$('#uploadForm').on('submit', function(event) {
    event.preventDefault();

    // Validate file input
    const fileInput = $('#image')[0];
    if (!fileInput.files.length) {
        $('#response').html(`
            <div class="error-message">
                Please select an image file.
            </div>
        `);
        return;
    }

    const file = fileInput.files[0];
    // Check file size (5MB limit)
    if (file.size > 5 * 1024 * 1024) {
        $('#response').html(`
            <div class="error-message">
                File size too large. Maximum size is 5MB.
            </div>
        `);
        return;
    }

    // Show loading indicator
    $('#loading').show();
    $('#submitBtn').prop('disabled', true);
    $('#response').empty();
    $('#analysis').empty();

    var formData = new FormData(this);
    $.ajax({
        url: 'https://www.bmrededucation.com/process.php',
        type: 'POST',
        data: formData,
        success: function(response) {
            if (response.status === 'success') {
                $('#response').html(`
                    <div class="success-message">
                        ${response.message}
                        ${response.data && response.data.full_url ?
                            `<br><small>File URL: <a href="${response.data.full_url}"
target="_blank">${response.data.full_url}</a></small>`
                            : ""
                        }
                    </div>
                `);

                if (response.data && response.data.full_url) {
                    analyzeImage(response.data.full_url);
                }
            } else {
```

```

        $('#response').html(
            <div class="error-message">
                ${response.message}
            </div>
        ');
    },
    error: function(xhr, status, error) {
        $('#response').html(
            <div class="error-message">
                Upload failed: ${error || 'Unknown error occurred'}
            </div>
        );
    },
    complete: function() {
        $('#loading').hide();
        $('#submitBtn').prop('disabled', false);
    },
    cache: false,
    contentType: false,
    processData: false,
    timeout: 30000 // 30 second timeout
    });
});

// Function to analyze the uploaded image
function analyzeImage(imageUrl) {
    $('#analysis').html(
        <div class="analysis-result">
            <div class="loading-spinner"></div>
            <p style="text-align: center; margin-top: 10px;">Analyzing image...</p>
        </div>
    );

    $.ajax({
        url: 'https://www.bmrededucation.com/proces2s.php',
        type: 'POST',
        data: JSON.stringify({ image_url: imageUrl }),
        contentType: 'application/json',
        success: function(response) {
            if (response.status === 'success') {
                try {
                    // Convert markdown to HTML if needed
                    const analysisHtml = marked.parse(response.data.analysis);
                    $('#analysis').html(
                        <div class="analysis-result">
                            <h3>Analysis Results</h3>
                            <div class="analysis-content">
                                ${analysisHtml}
                            </div>
                        </div>
                    );
                } catch (e) {
                    // Fallback to plain text if markdown parsing fails
                    $('#analysis').html(

```

```

        <div class="analysis-result">
            <h3>Analysis Results</h3>
            <div class="analysis-content">
                ${response.data.analysis}
            </div>
        </div>
    `);
    }
    } else {
        $('#analysis').html(`
            <div class="error-message">
                Analysis failed: ${response.message}
            </div>
        `);
    }
    },
    error: function(xhr, status, error) {
        $('#analysis').html(`
            <div class="error-message">
                Analysis error: ${error || 'Failed to analyze image'}
            </div>
        `);
    }
    });
}

// Add this CSS for the loading spinner
const spinnerCSS = `
    <style>
        .loading-spinner {
            width: 50px;
            height: 50px;
            border: 4px solid #e2e8f0;
            border-top: 4px solid #4299e1;
            border-radius: 50%;
            animation: spin 1s linear infinite;
            margin: 0 auto;
        }

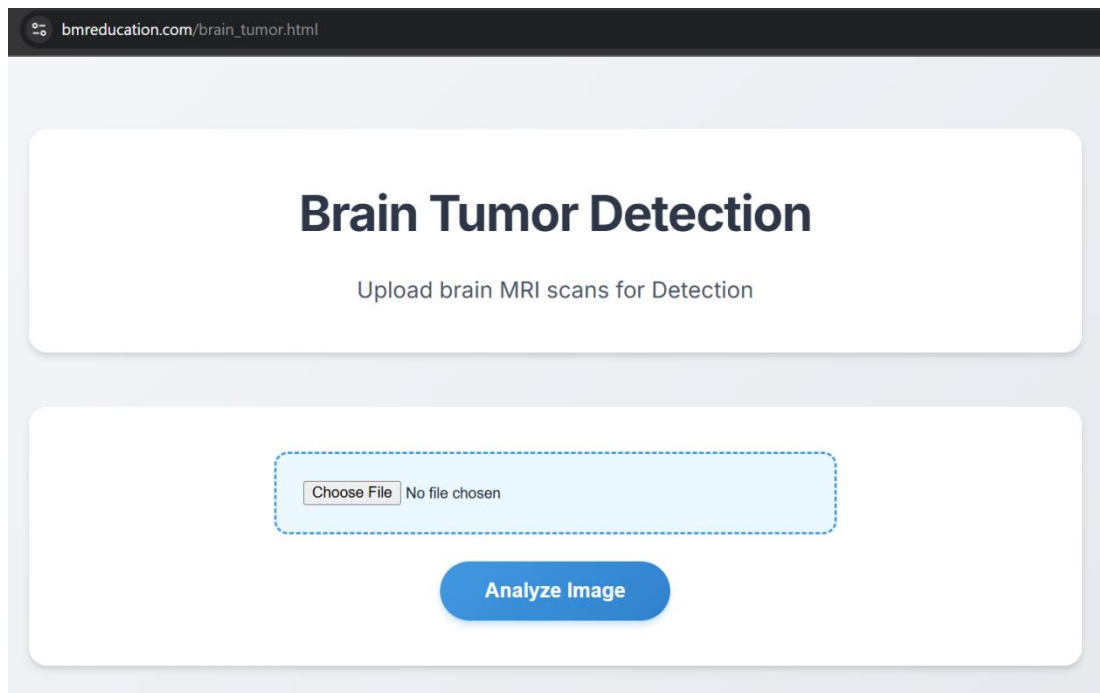
        @keyframes spin {
            0% { transform: rotate(0deg); }
            100% { transform: rotate(360deg); }
        }
    </style>
`;

// Append the spinner CSS to the head
$('#head').append(spinnerCSS);
</script>
</body>

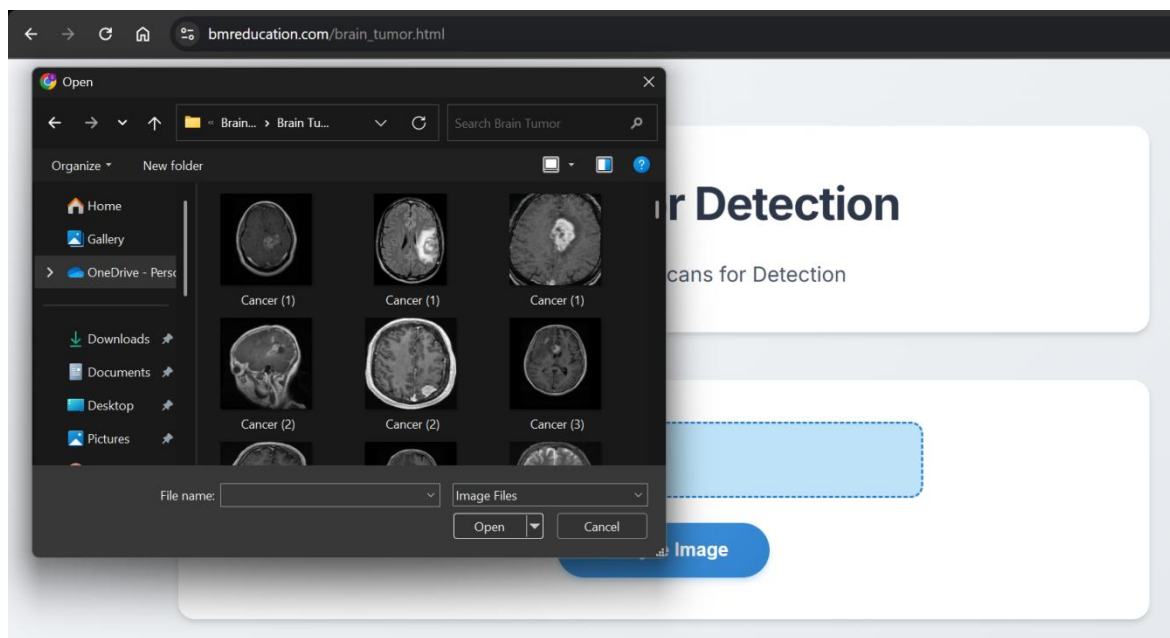
```

APPENDIX-B

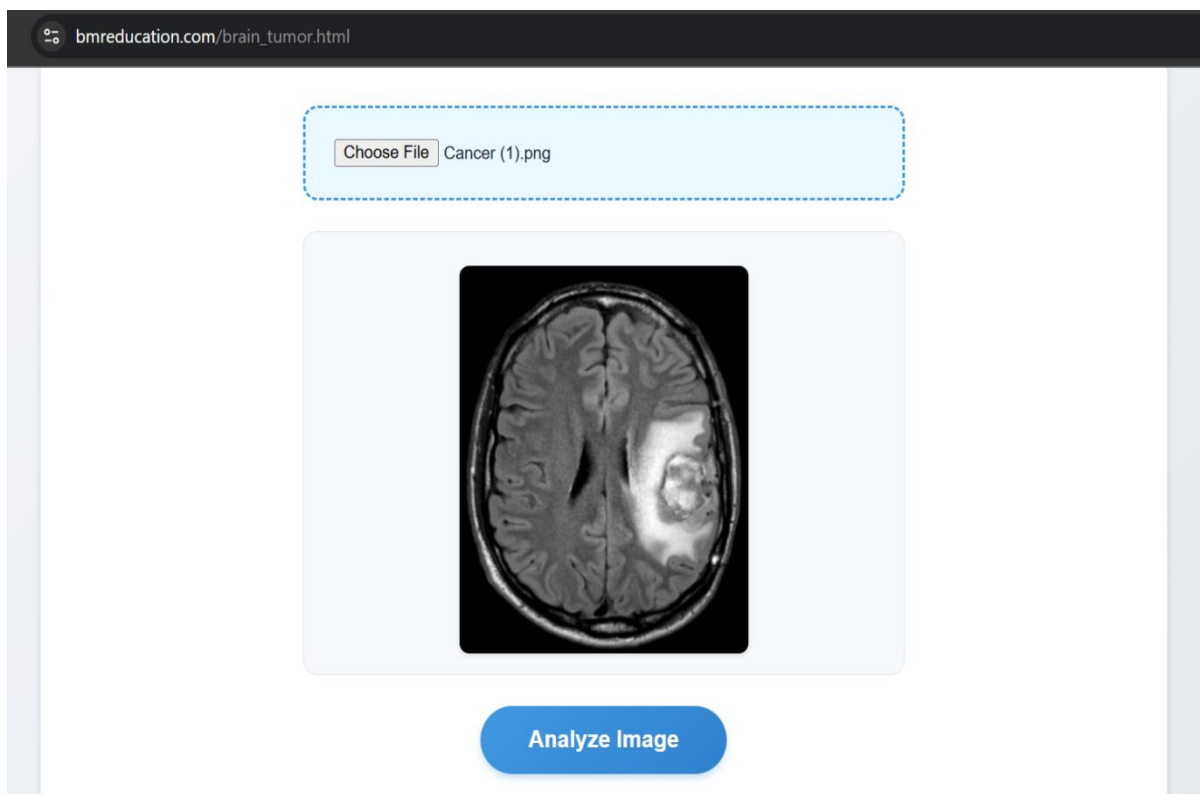
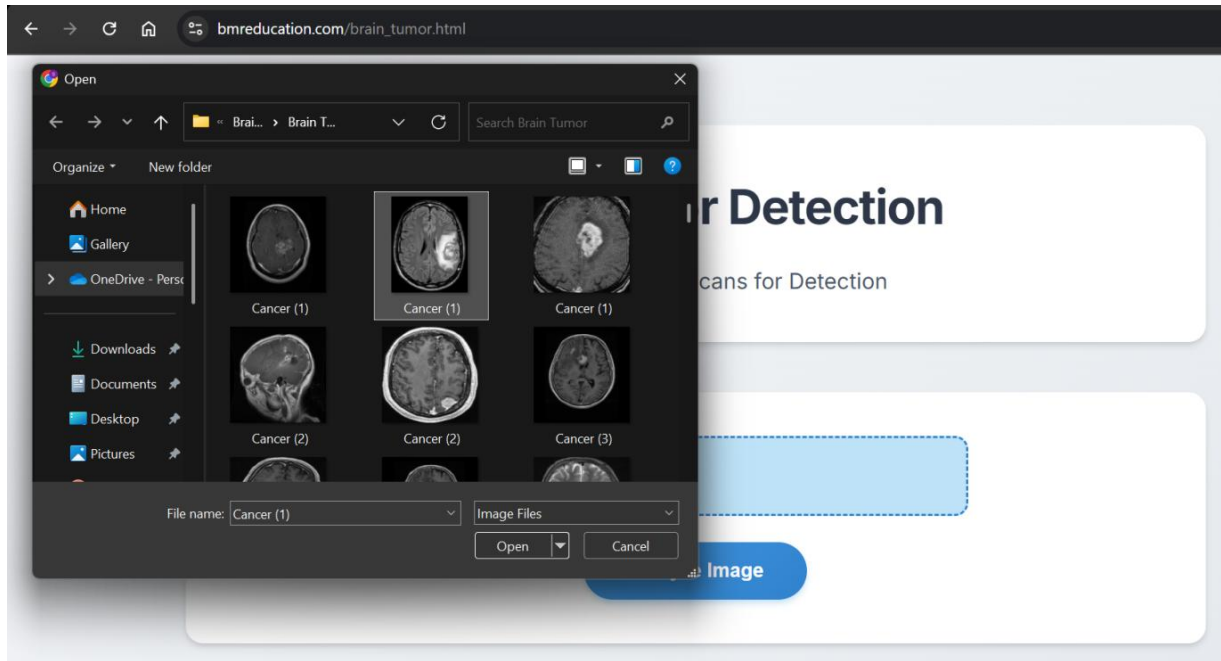
SCREENSHOTS



- The webpage asks the user to select an image for analysis



- The user can choose an image from any dataset which contains healthy or tumour present MRI scans.



- The user uploads the image and waits for the analysis.

- Analysis Result:

 bmreducation.com/brain_tumor.html

Analysis Results

[TUMOR_DETECTION]

- Status: Present
- Location: Right parietal lobe
- Size: Approximately 2.5 cm in diameter
- Shape: Irregularly rounded
- Margins: Ill-defined, infiltrative

[TECHNICAL_METRICS]

 bmreducation.com/brain_tumor.html

[TECHNICAL_METRICS]

- Detection Confidence: 95%
- Key Measurements: Tumor diameter: 25mm; Area: 5.0 cm² (approximate)

[BRAIN_ASSESSMENT]

- Structure Status: Abnormal due to the presence of a mass in the right parietal lobe causing mass effect.
- Ventricle Analysis: Mild shift of midline structures to the left, suggesting mass effect. Right lateral ventricle may be slightly compressed.
- Matter Assessment: Gray and white matter appear relatively normal in the unaffected regions. The mass shows altered signal intensity compared to surrounding brain tissue.
- Secondary Findings: Mild midline shift.

 bmreducation.com/brain_tumor.html

- Matter Assessment: Gray and white matter appear relatively normal in the unaffected regions. The mass shows altered signal intensity compared to surrounding brain tissue.
- Secondary Findings: Mild midline shift.

[ML_ANALYSIS]

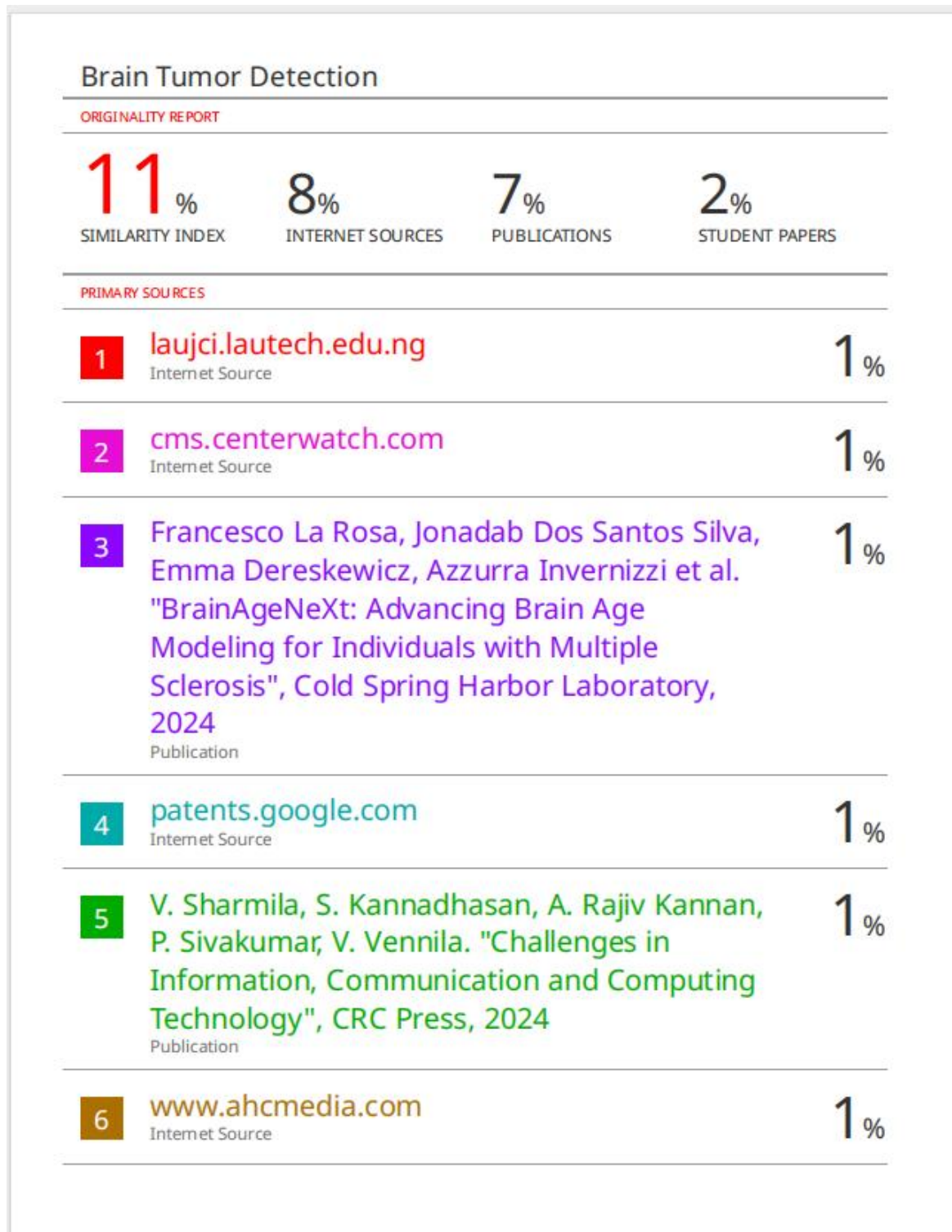
- Top Diagnoses: Glioblastoma (75%), Meningioma (15%), Metastatic Tumor (10%)
- Regional Scores: Right Parietal Lobe: 98%; Left Parietal Lobe: 90%; Frontal Lobes: 85%; Temporal Lobes: 80%; Cerebellum: 70%

APPENDIX-C

ENCLOSURES

1. Paper if formulated and Communicated.
2. Similarity Index / Plagiarism Check report clearly showing the Percentage (%). No need for a page-wise explanation.

PLAGARISM REPORT OF PUBLISHED RESEARCH PAPER



PLAGARISM REPORT OF PROJECT REPORT

Reportttt

ORIGINALITY REPORT

19%	12%	13%	9%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to University of Essex Student Paper	1%
2	Dinesh Goyal, Bhanu Pratap, Sandeep Gupta, Saurabh Raj, Rekha Rani Agrawal, Indra Kishor. "Recent Advances in Sciences, Engineering, Information Technology & Management - Proceedings of the 6th International Conference "Convergence2024" Recent Advances in Sciences, Engineering, Information Technology & Management, April 24-25, 2024, Jaipur, India", CRC Press, 2025 Publication	1%
3	Biswadip Basu Mallik, Gunjan Mukherjee, Rahul Kar, Aryan Chaudhary. "Deep Learning Concepts in Operations Research", Routledge, 2024 Publication	1%
4	fastercapital.com Internet Source	1%
5	discuss.logseq.com Internet Source	1%

MAPPING THE PROJECT TO SDG MAPPING



The Project work carried out here is mapped to SDG-3 Good Health and Well-Being.

The primary goal of brain tumor detection through deep learning is to improve healthcare quality, reduce diagnosis time, and increase survival rates by providing more accurate and early diagnoses.

PUBLICATION CERTIFICATES



**International Journal of
Innovative Research in Technology**
An International Open Access Journal Peer-reviewed, Refereed Journal
www.ijirt.org | editor@ijirt.org An International Scholarly Indexed Journal

Certificate of Publication

The Board of International Journal of Innovative Research in Technology
(ISSN 2349-6002) is hereby awarding this certificate to

SEETHA THANOOJA K

In recognition of the publication of the paper entitled

**DETECTION OF BRAIN TUMOURS FROM MRI IMAGES USING
CONVOLUTIONAL NEURAL NETWORKS (CNNs)**

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor
Published in Volume 11 Issue 8, January 2025
Registration ID 171850 Research paper weblink: <https://ijirt.org/Article?manuscript=171850>


EDITOR


EDITOR IN CHIEF


ISSN 2349-6002
9 772349 600209





**International Journal of
Innovative Research in Technology**
An International Open Access Journal Peer-reviewed, Refereed Journal
www.ijirt.org | editor@ijirt.org An International Scholarly Indexed Journal

Certificate of Publication

The Board of International Journal of Innovative Research in Technology
(ISSN 2349-6002) is hereby awarding this certificate to

RAMAMURTHY KETHA

In recognition of the publication of the paper entitled

**DETECTION OF BRAIN TUMOURS FROM MRI IMAGES USING
CONVOLUTIONAL NEURAL NETWORKS (CNNs)**

Published in IJIRT (www.ijirt.org) ISSN UGC Approved (Journal No: 47859) & 7.37 Impact Factor
Published in Volume 11 Issue 8, January 2025
Registration ID 171850 Research paper weblink: <https://ijirt.org/Article?manuscript=171850>


EDITOR


EDITOR IN CHIEF


ISSN 2349-6002
9 772349 600209



